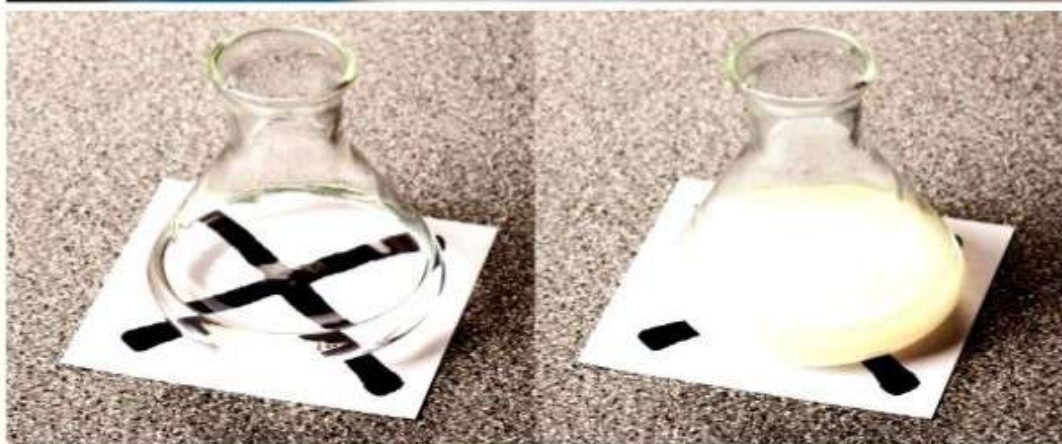


CHEMISTRY PRACTICAL FOR ORDINARY LEVEL

FORM 3 & 4



Iron(II)
Chloride
 FeCl_2



Magnesium
Chloride
 MgCl_2



Zinc
Chloride
 ZnCl_2



Sodium
chloride,
 NaCl



Copper(II)
Chloride
 CuCl_2



Copper(II)
sulphate
 CuSO_4



Aluminium(II)
sulphate
 $\text{Al}_2(\text{SO}_4)_3$



Potassium
chloride,
 KCl



Lukago,D

CHEMISTRY

PRACTICAL

FOR ORDINARY LEVEL STUDENT'S BOOK

Daudi Lukago 2023

Published 2023

©Daudi Lukago

ISBN: :9798600861749

Mobile numbers; **+255758821626**

+255625605414

Email; **ntungi1000@gmail.com**

Dar es salaam, Tanzania

All rights reserved. No part of this textbook may be reproduced, stored in any retrieval system or transmitted in any form or by any means whether electronic, mechanical, photocopying, recording or otherwise, without the prior written permission of the Tanzania Institute of Education

Table of Contents

CHAPTER 1.....	4
VOLUMETRIC ANALYSIS.....	4
Basic principles of volumetric analysis.....	4
Importance of (volumetric) quantitative analysis.....	4
Molarity	4
Concentration.....	5
Standard Volumetric Apparatus.....	5
Titration.....	10
Condition suitable for Titration.....	11
Common acid-base indicator.	11
Physical properties of an acid.....	12
Basicity of an acid is the number of replaceable hydrogens.....	12
Physical properties of base.	12
Neutralization reaction is the reaction between an acid and a base to produce a salt and water.....	13
Common acid-base indicators.....	13
Precautions during titration.	13
TITRATION CAN BE USED TO DETERMINE.....	14
2.Percentage purity of a substance.....	14
3.Amount of water of crystallization in a substance.....	15
Application of volumetric analysis.	15
PRACTICAL QUESTIONS AND THEIR ANSWERS	16
.....	23
.....	25
.....	26

.....	26
CHAPTER 2.....	60
CHEMICAL KINETICS AND EQUILIBRIUM	60
Factors affecting the rate of chemical reaction.	60
EFFECT OF CONCETRATION OF REACTANTS ON THE RATE OF CHEMICAL REACTION.	60
Chemicals used in this practical	60
Effect Of Temperature on The Rate Chemical Reaction	63
Uses of Sulphur.....	65
All Actual Practical questions of Concentration and Temperature.....	Error!
Bookmark not defined.	
Chapter 3	83
QUALITATIVE ANALYSIS.....	83
Importance of qualitative analysis.	84
Types of qualitative analysis.....	84
Physical properties of salts.....	84
Effect of heat on a salt.	85
Flame tests	86
Stock solution is a large volume of common reagents such as NaOH or NH ₄ OH at a standardized concentration from which one can dilute into a working concentration.	86
QUALITATIVE ANALYSIS PROCEDURES.	88
CONFIRMATORY TEST OF ANIONS	88
CONFIRMATORY TEST OF CATIONS.....	89
Uses of Ammonia	89
Uses of Calcium and its Compounds.	90
Uses of Sodium and Its Compounds.....	90
Uses of Copper and its compounds.....	90

Uses of Iron and its compounds.....	91
Uses of Zinc and its compounds.....	91
Uses of Lead and its compounds.....	91
CHAPTER 4.....	120
PREPARATION OF STANDARD ACID SOLUTION AND DILUTION FORMULA	120
APPENDIX 1:	123
Appendix 2: Periodic table of the elements.....	124
Reference books.....	125

CHAPTER 1.

VOLUMETRIC ANALYSIS.

Volumetric analysis

Volumetric analysis is the determination of the concentration of an analyte in a solution by titration (procedures which are used to measure the volume of reagent needed to react with the analyte).

It is also called titrimetric analysis. In titration, the reagent of a given concentration is called a titrant. The titrant is added drop-wise to the analyte until the reaction is complete. The titrant should not be in excess, but in an exact amount necessary for stoichiometric reaction with the analyte. The quantity of the titrant required for complete reaction with the analyte is used to calculate the quantity of the analyte present in the sample. The most common titrations are acid-base, oxidation-reduction (Redox), complex formation and precipitation titrations.

Basic principles of volumetric analysis.

- ❖ The one solution to be analyzed contains unknown amount of chemicals
- ❖ The reagent of known concentration reacts with a chemical of unknown amount in the presence of indicator which shows the end-point.
- ❖ The amount of unknown chemical in the measured volume of solution is calculated by using the mole ratio from the reaction equation.

Importance of (volumetric) quantitative analysis

- ❖ It helps to determine volumes and concentration of reacting solutions.

A known volume of a solution of unknown concentration is reacted with a known volume of a solution of known concentration.

The standard solution is usually delivered from a burette so the volume added is measured.

Standard solution is a solution of known concentration(molarity).

Standardization is the procedure used to determine the concentration of a solution (analyte)

Molarity

By definition: Molarity is the concentration of a solution expressed in moles per litre. Also, it called molar solution. SI Unit of molarity is mol/dm³ or M

Mathematically

$$\text{Molarity} = \frac{\text{concentration}}{\text{molar mass}}$$

Concentration.

By definition: Concentration is the quantity of the solute in the solution. SI Unit of concentration is g/dm³

Mathematically

$$\text{Concentration} = \text{molarity} \times \text{molar mass}.$$

Standard Volumetric Apparatus.

The apparatus that are needed in volumetric analysis depends on complexity of the work.

The following are apparatus used in volumetric analysis practical;

- ❖ Burette
- ❖ Pipette
- ❖ Volumetric flasks
- ❖ Conical flask
- ❖ Filter funnel
- ❖ Retort stand with clamp
- ❖ White tile or paper
- ❖ Measuring cylinders
- ❖ Wash bottle
- ❖ Beakers
- ❖ Dropper

Burette

Burette is a graduated tube with one open end and a stop tap at the other end. It is used to measure accurate volume of the titrant (acid) during titration.

The solution added to the burette is called **titrant**



Note:

- ❖ Common burette has a capacity of 50 mls (cm^3)
- ❖ Reading of burette should be in two decimal places

Note:

- ✓ Initial reading should not necessary be zero
- ✓ Hold a black-stripped white card behind the meniscus when taking the reading in order to ensure the meniscus is clearly visible
- ✓ Solution filled in burette is called titrant
- ✓ View the meniscus horizontal to obtain correct reading which tend to avoid parallax error
- ✓ During titration top of burette should be centre and dipped into the flask

Pipette.

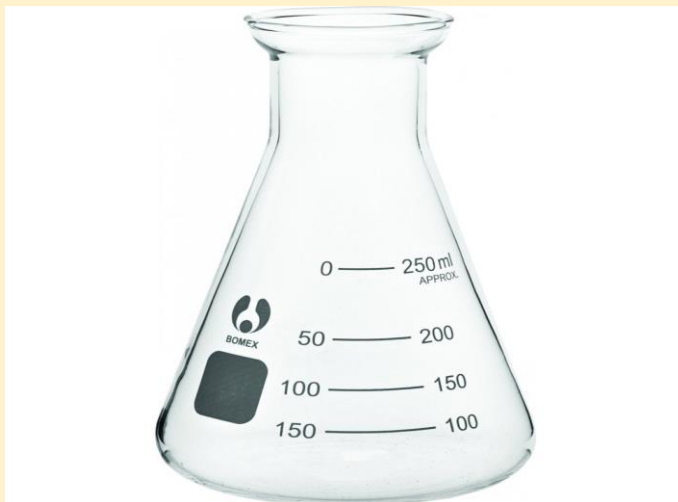
Pipette is a thin glass tube with a wide at the middle part. It used to measure and transfer definite volume of base into a conical flask. Common pipette used in chemistry practical has a capacity of (20 or 25) cm^3



Conical flask.

- ❖ It is used to mix solutions during titrations.
- ❖ It is conical in shape and ensures the solutions does not spill out as you swirl the flask.

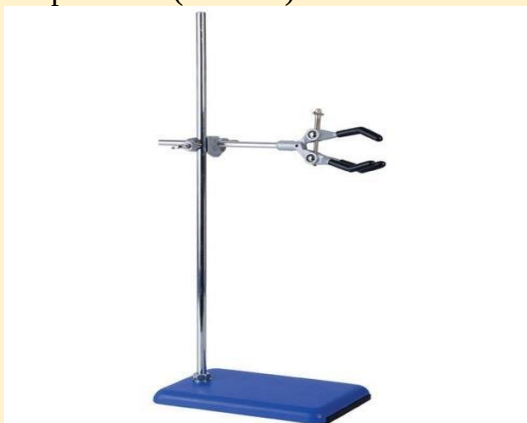
Conical flask mostly used has capacity of 250 cm³ which has a neck which enable easily held while swirling the contents inside.



Note: it should be rinsed with distilled water before using it.

Retort stand and clamp.

It used to hold the burette in an upright position while performing volumetric analysis experiment (titration).



White tile or paper.

It is kept under conical flask to give a clear background for observing colour change at end point.

**Measuring cylinders.**

It is used to measure volume of reagents (titrant and Analyte) required during volumetric analysis experiment.

**Beaker.**

It is used to store (hold) reagents (titrant and Analyte) during volumetric analysis experiment.



Dropper.

It used to add the indicator into the solution inside conical flask.

**Plastic wash bottle.**

Used to store distilled water.

**Volumetric flasks.**

Volumetric flask is flask used to prepare standard solution from stock solution. It calibrated to contain specific volume of solution. It has different capacities ranging (250 – 500) cm^3 . It has graduated mark at its neck.



Filter Funnel.

It used to pour a solution into the burette without spilling solution.



Titration.

Titration involves the technique of adding standard solution (known solution) from burette to flask (unknown solution) .

Note:

- ❖ Known solution is also called titrant
- ❖ Unknown solution is also known as Analyte

Condition suitable for Titration.

- ❖ Reaction must be fast.
- ❖ Reaction must be irreversible.
- ❖ Reaction must be represented by a single chemical equation.
- ❖ Reaction must have definite end point.

Note: only Acid-base titration will be discussed at this level.

Acid-Base titration.

Acid-Base involves the technique of adding standard solution (Acid) from burette to flask (base) and vice versa until the end point is indicated by indicator by changing colour of the mixture solution.

Common acid-base indicator.

Indicator which are most used to change colour at end point are;

- ❖ Methyl orange and
- ❖ Phenolphthalein

Detection of the end point

The end point of a titration is detected visually by a sudden change in some properties of a reaction mixture, in many cases by change in colour.

In acid-base titrations, both the acid and base are always colourless, so it is difficult to tell when the reaction between acid and base has completed.

An indicator is added to the analyte solution in order to detect the end point of the reaction by following the colour change of the indicator. For example, in titration of HCl solution with NaOH solution, methyl orange or phenolphthalein are commonly used as indicators.

- ❖ Methyl orange has a yellowish colour in NaOH (basic) solution. After the completion of the reaction (*i.e.* after equivalence point), an extra drop of HCl solution turns the reaction mixture orange.
- ❖ Phenolphthalein indicator is pink in NaOH so after equivalence point additional drop of HCl will turn the reaction mixture colourless.

5.1.4 Acid-base indicators

The success of any titration process depends upon how accurately the equivalence point is judged.

Indicators are substances which at equivalence point produces some changes which can be readily

seen. If the end point coincides with the equivalence point, it is an ideal condition.

In acid-base titrations, both the acid and base are always colourless, so it is difficult

to tell when the reaction between acid and base

An Acid is chemical substance which when dissolved in water produces hydrogen ions(H^+) as the only positive ions.

Physical properties of an acid.

- ❖ They have sour taste.
- ❖ Turn blue litmus paper red.
- ❖ Acids have a corrosive action.

Strong acids-is an acid which dissociate completely to give hydrogen ions(H^+) in water.

Examples of strong acids.

- ❖ Sulphuric acid (H_2SO_4)
- ❖ Hydrochloric acid(HCl)
- ❖ Nitric acid(HNO_3)

Weak acid- is an acid which dissociate partially to give hydrogen ions(H^+) in water.

Example of weak.

- Ethanoic acid (CH_3COOH)
- Oxalic acid ($HOOC-COOH$) or $(COOH)_2$

Basicity of an acid is the number of replaceable hydrogens

Base is an oxide or hydroxide of metal which reacts with an acid to form salts and water only.

Physical properties of base.

- ❖ They have bitter taste
- ❖ They turn red litmus paper blue
- ❖ They have slippery soapy feeling
- ❖ Mainly are oxides and hydroxides of metals

Strong base.is an acid which dissociate completely to give hydroxyl ions(OH^-) in water.

Examples of strong base

- ❖ Potassium hydroxide(KOH)
- ❖ Sodium hydroxide($NaOH$)
- ❖ Sodium carbonate(Na_2CO_3)

Weak base. is an acid which dissociate partially to give hydroxyl ions(OH^-) in water. For example

- Ammonium hydroxide(NH_4OH)

Neutralization reaction is the reaction between an acid and a base to produce a salt and water.

Indicators are compounds that show definite colour change when in an acid or base.

Choice of indicator to be used on titration depends on the properties (strength) of the reactants.

Common acid-base indicators

Indicator	Colour in acidic medium	Colour in basic medium
Methyl orange(MO)	Pink/red	Yellow
Phenolphthalein(POP)	Colourless	Pink/violet

Precautions during titration.

- ❖ Avoid contamination
- ❖ view meniscus level on the burette with an eye at horizontal level

Example of acid-base titration and their suitable indicator to be used

Acid-Base Titration	Example	Suitable Indicator
Strong acid vs strong base	H_2SO_4 vs NaOH	Any indicator
Strong acid vs weak base	HCl vs NH_4OH	Methyl orange
Weak acid and strong base	$\text{C}_2\text{H}_4\text{O}_2$ vs KOH	POP
Weak acid vs weak base	$\text{H}_2\text{C}_2\text{O}_4$ vs NH_4OH	No suitable indicator

NOTE

The results obtained during experiment is recorded in tabular form with two decimal places as shown in the table below.

Burette reading	Pilot	1	2	3
Final reading(cm^3)	V			
Initial reading(cm^3)	V_0			
Volume of acid used(cm^3)	$V - V_0$			

TITRATION CAN BE USED TO DETERMINE.

1.Relative atomic mass of unknown element.

Relative atomic mass of an unknown element in an acid or base can be determined by titrating standard solution of acid or base.

procedures to be followed.

1. Calculate the average volume of unknown acid.
2. Write the balanced chemical for the neutralization reaction.
3. Calculate the molarity of unknown acid or base by using titration formula.

$$\frac{MaVa}{MbVb} = \frac{na}{nb}$$

Where;

Ma –molarity of an acid

Va –volume of an acid

na-number of mole(s) of an acid

Mb-molarity of base

Vb-volume of base

nb –number of moles of base

NOTE

na and *nb* are obtained from a balanced chemical equation.

4. Calculate the concentration of unknown acid or alkali(g/dm³) if not given

$$\text{Concentration} = \frac{\text{mass}}{\text{volume}}$$

5. Calculate the molar mass of the unknown acid or base

$$\text{Molar mass} = \frac{\text{concentration}}{\text{molarity}}$$

6. Determine the relative molecular mass of the unknown acid or base

2.Percentage purity of a substance.

Percentage purity of a substance is the percentage of pure substance in a sample

$$\text{Percentage purity} = \frac{\text{concentration of pure substance}}{\text{concentration of impure substance}} \times 100\%$$

Procedures to be followed.

- i. Calculate the average volume of the acid from the table of results

- ii. Write the balanced chemical equation to get mole ratio (na and nb)
- iii. Calculate the concentration of pure substance (acid or base)
- iv. Workout for the percentage purity of the sample asked.

3.Amount of water of crystallization in a substance.

Water of crystallization is the water that is bound within the crystals of a substance.

Procedures to be followed.

- 1) Calculating the average volume of an acid
- 2) Write the balanced chemical equation to get mole ratio
- 3) Determine the molar concentration of the substance (acid or base)
- 4) Calculate the relative molecular mass of the substance (acid or base)

$$\text{Molecular mass} = \frac{\text{concentration}}{\text{molarity}}$$

(v)Determine the number of molecules of water of crystallization in the sample

Application of volumetric analysis.

In different fields volumetric analysis are used as follows

1. Medicine.

- ❖ Used to determine concentration of virus/bacteria in a blood sample.
- ❖ Used to determine amount of acid which can be neutralized by antacid tablet.

2. Industries.

- ❖ Used to determine amount of acetic acid in vinegar.
- ❖ Used to analyze the acidity of fruit juices.
- ❖ Used to analyze the household ammonia (used as a cleaning fluid).

3. Analysis of water.

- ❖ Used to determine hardness of water

4. Agriculture.

- ❖ Used to determine composition of soil substances contained, Substance include nitrogen, potassium and phosphorous.

5. Other application.

- ❖ Used to determine percentage of iron in iron ore

- ❖ Used to determine salt content in brine (water containing dissolved salt)
- ❖ Used to determine percentage mass of copper in a copper salt

PRACTICAL QUESTIONS AND THEIR ANSWERS

NECTA 2022

CHEMISTRY 2B

1. you are provided with solutions **BB** and **CC**. one of these solutions is acidic and the other is basic. The acidic solution contains 3.06 g/dm^3 sulphuric acid (H_2SO_4). the basic solution was made by dissolving 2g of an impure sodium hydroxide (NaOH) in distilled water to make a 500cm^3 of a solution

Procedure

- (i) Pour about 2cm^3 of solution **BB** into a test tube and use a litmus paper to test if it is acidic or basic
- (ii) Discard the content and wash the test tube.
- (iii) Repeat the step(i) and (ii) using solution **CC**.
- (iv) Titrate the acid solution (in the burette) against the base solution (in the titration flask) using methyl orange (**MO**) as an indicator up to the end point
- (v) Repeat the step (iv) to obtain three more readings and record your results in a tabular form

Questions

- a) (i) what was the volume of pipette used
- (ii) Calculate the average volume of acid used.
- (iii) What was the colour change at end point?

- b) Write the balanced chemical equation for the reaction for the reaction between solution BB and CC
- c) Calculate the percentage impurity of sodium hydroxide.

Solution

(a)(i) the volume of pipette used was 25 cm³

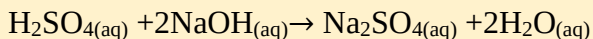
Solution BB was acid the pink colour was observed after addition of methyl orange(MO)

Burette reading	Pilot	1	2	3
Final reading(cm ³)	40.70	40.50	40.60	40.50
Initial reading(cm ³)	0.00	0.00	0.00	0.00
Volume of acid used(cm ³)	40.70	40.50	40.60	40.50

$$\begin{aligned}
 \text{(ii) Average volume of BB} &= \frac{V_1 + V_2 + V_3}{3} \\
 &= \frac{40.50 + 40.60 + 40.50}{3} \\
 &= 40.50 \text{ cm}^3
 \end{aligned}$$

(iii) colour change was from yellow to pink colour.

(b) Balanced chemical equation for the reaction taking place between BB and CC



(c) Data

Molarity of acid (ma)=0.03122M

Volume of acid(va)= 40.00cm³

Volume of base(vb)=25cm³

Number of moles of base(nb)=2

Number of moles of acid(n_a)=1

Molarity of base (M_b)=?

From titration formula.

$$\frac{m_a v_a}{m_b v_b} = \frac{n_a}{n_b}$$

$$\frac{0.03122 \times 40.00 \times 2}{25 \times 1} = 0.099904M$$

Molarity of base (NaOH) was 0. 099904M.

Concentration of pure NaOH=*molarity X molar mass*

Molar mass of NaOH= 23+16+1=40g/mol

Concentration of pure NaOH=0. 099904M $\times$ 40g/mol=3.99616g/dm³.

Concentration of impure NaOH= $\frac{\text{mass}}{\text{volume}} = \frac{2g}{0.5dm^3} = 4g/dm^3$

Percentage purity = $\frac{\text{conc of pure}}{\text{conc of impure}} \times 100\%$

Percentage purity= $\frac{3.9961}{4} \times 100\% = 99.9025\%$

Percentage purity of NaOH was 99.9025%

Percentage impurity=100% - percentage purity

Percentage impurity=100% -99.9025%=0.0975%

Percentage impurity NaOH was 0.0975%

CHEMISTRY 2A

1. You are provided with the following solution

LL: containing 6.3g of dibasic acid, $\text{H}_2\text{C}_2\text{O}_4 \cdot \text{XH}_2\text{O}$ dissolved to make 1 litre of solution and

MM: containing aqueous sodium hydroxide made by dissolving 4.0 g of the salt dissolved to make 1 litre of solution.

Determine the value of X in the acid $\text{H}_2\text{C}_2\text{O}_4 \cdot \text{XH}_2\text{O}$

- (i) Fill the burette with solution LL
- (ii) Pipette 20/25cm³ of solution MM and put in a conical flask
- (iii) Titrate LL against MM using two drops of POP indicator
- (iv) Repeat the titration to obtain three more titre value and record in tabular form.

Questions

- a)
 - (i) Indicate the volume of pipette used.
 - (ii) Complete the table of results and compute the average volume of acid used for complete neutralization of MM.
- b) Calculate the molarity of the base
- c) Write a balanced chemical equation for the reaction taking place
- d) Calculate the molarity of the acid
- e) Calculate
 - (i) The value of X in $\text{H}_2\text{C}_2\text{O}_4 \cdot \text{XH}_2\text{O}$
 - (ii) The percentage of water of crystallization in $\text{H}_2\text{C}_2\text{O}_4 \cdot \text{XH}_2\text{O}$

Solution

- a)
 - (i) The volume of pipette used was 25cm³
 - (ii) Table of results

Burette reading	Pilot	1	2	3
Final reading (cm ³)	25.00	24.90	25.00	25.00
Initial reading (cm ³)	0.00	0.00	0.00	0.00
Volume used (cm ³)	25.00	25.90	25.00	20.00

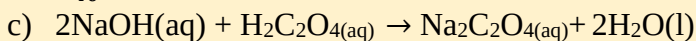
$$\text{Average volume of acid (E)} = \frac{V_1 + V_2 + V_3}{3}$$

$$V_a = \frac{24.90 + 25.00 + 25.00}{3}$$

$$V_a = \underline{25.00 \text{ cm}^3}$$

$$\text{b) Molarity of base} = \frac{\text{concentration g dm}^{-3}}{\text{molar mass g mol}^{-1}}$$

$$= \frac{4}{40} = 0.1 \text{ M}$$



d) calculate molarity of acid H₂C₂O₄

From the relation

$$\frac{m_a v_a}{m_b v_b} = \frac{n_a}{n_b}$$

Where:

Ma = Molarity of Acid (LL)

Va = Volume of Acid (LL)

Mb = Molarity of Base (MM)

Vb = Volume of Base (MM)

na = Number of moles of Acid (LL)

nb = Number of moles of Base (MM)

from the balanced chemical equation

$$n_a = 1 \quad n_b = 2$$

We have

$$18x=126-90$$

$$X=\frac{36}{18}=2$$

(ii) Percentage of water crystallization in $H_2C_2O_2 \cdot xH_2O$

$$=\frac{36}{126} \times 100\% = 28.57\%$$

PRACTICAL ACTIVITY

1. You are provided with the following

F: A solution made by diluting 50 ml of 1.5M oxalic acid ($H_2C_2O_4$) to make 5litres of a solution.

G: A solution containing 0.45g of impure sodium hydroxide ($NaOH$) in 0.5litre of a solution.

MO Methyl orange indicator

POP Phenolphthalein indicator

Procedure

- (i) Pour solution **F** in the burette
- (ii) Pipette 20ml or 25ml of solution **G** and put it in the titrating flask then add two drops of the suitable indicator among the indicators given above.
- (iii) Titrate solution **F** in the burette against solution **G** in the flask until the colour change is observed. Record the volume of solution F required to neutralize solution G.
- (iv) Repeat the procedure (i) to (iv) to obtain three more volumes of solution F. Tabulate your results in table format.
 - a) Find the average volume used.
 - b) (i) _____ ml of solution **F** required _____ ml of solution **G** for complete neutralization.
 - (ii) The colour of the indicator at the end point was _____.
 - c) Which indicator did you choose? Why?

d) How many moles of solution **F** were required for complete reaction?

e) Showing your calculations clearly, calculate the percentage purity of sodium hydroxide.

f) Suggest two sources of error in this experiment.

Solution

a) Table of results

Titration	pilot	1	2	3
Final volume (ml)	15.40	35.10	15.00	34.90
Initial volume (ml)	0.00	20.00	0.00	15.00
Volume used (ml)	15.40	15.10	15.00	14.90

b) Required: Average volume used,

$$V_a = \frac{V_1 + V_2 + V_3}{3}$$

$$V_a = \frac{(15.10 + 15.00 + 14.90)ml}{3}$$

$$V_a = 15.00 \text{ ml}$$

The average volume used was 15.00 ml

c) (i) 15 ml of solution **F** required 25 ml of solution **G** for complete neutralization

(ii) The colour of the indicator at the end point was from pink to colourless.

d) Phenolphthalein indicator.

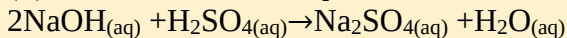
Reason: Because the titration was between weak acid and strong base.

e) Required: Moles of solution **F** required for complete reaction.

$$\text{Average volume of acid used} = \frac{v_1 + v_2 + v_3}{3} = \frac{26.00 + 26.10 + 26.00}{3}$$

$$\text{Average volume} = 26.00 \text{ cm}^3.$$

- (c) (i) 25 cm^3 of **P** required 26.00 cm^3 of **Q** for complete reaction
 (ii) balanced chemical equation between **P** and **Q**



(d) Molarity of base = $\frac{\text{concentration of base}}{\text{molar mass of base}}$

$$\text{Concentration} = \frac{\text{mass}}{\text{volume}} = \frac{2 \text{ g}}{0.5 \text{ dm}^3} = 4 \text{ g/dm}^3$$

$$\text{Molar mass of NaOH} = 23 + 16 + 1 = 40 \text{ g/mol}$$

$$\text{Molarity of base (mb)} = \frac{4 \text{ g/dm}^3}{40 \text{ g/mol}} = 0.1 \text{ M}$$

$$\text{Volume of base (vb)} = 25 \text{ cm}^3$$

$$\text{Volume of acid (va)} = 26.00 \text{ cm}^3$$

$$\text{Number of moles of acid (na)} = 1$$

$$\text{Number of moles of base (nb)} = 2$$

$$\text{Molarity of acid (ma)} = ?$$

From titration formula

$$\frac{m_a v_a}{m_b v_b} = \frac{n_a}{n_b}$$

$$M_a = \frac{M_b v_b n_a}{v_a n_b}$$

$$M_a = \frac{0.1 \times 25 \times 1}{26.00 \times 2} = 0.05 \text{ M}$$

Molarity of acid was 0.05M

Concentration of pure sulphuric acid

From

$$\text{Molarity} = \frac{\text{concentration}}{\text{molar mass}}$$

$$\text{Molar mass of H}_2\text{SO}_4 = 1 \times 2 + 32 + 16 \times 4 = 98 \text{ g/mol}$$

$$\text{Concentration of pure} = \text{molarity} \times \text{molar mass}$$

$$= 0.05 \text{ M} \times 98 = 4.9 \text{ g/dm}^3$$

$$\text{Concentration of impure acid} = \frac{\text{mass}}{\text{volume}}$$

$$=5.2\text{g}/1\text{dm}^3$$

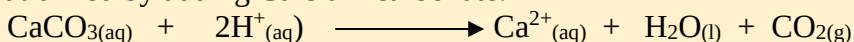
$$\text{Percentage purity} = \frac{\text{concentration of pure}}{\text{concentration of impure}} \times 100\%$$

$$\frac{4.9\text{g}/\text{dm}^3}{5.2\text{g}/\text{dm}^3} \times 100\% = 94.20\%$$

Percentage purity of sulphuric acid was 94.20%

PRACTICAL PROBLEM

1. A large volume of water has been contaminated with acid. Before the contaminated water can be disposed off, the acid can be neutralized by adding Calcium carbonate.



You are required to determine concentration of hydrogen ions present in the contaminated water and then calculate mass of calcium carbonate needed to neutralize all the acids.

Solution **R** is a sample of the contaminated water

Solution **L** is 0.1M Sodium hydroxide

Solution **MO** is Methyl Orange indicator.

Procedures

- i) Put R into the burette.
- ii) Pipette 25cm³ or 20cm³ portion of L into a flask and titrate with R, Using the **MO** provided
- iii) Perform three or more titrations and record the results in a tabular form.

Questions

- (a) Summary; ____cm³ of the average volume of **R** required ____cm³ of **L** for complete neutralization.
- (b) The colour change at the end point was from ____ to ____
- (c) Why the result of the first titration is usually ignored?
- (d) Calculate;
 - (i) The concentration of hydrogen ion in Moles/dm³
 - (ii) The number of moles of hydrogen ions present in 10,000cm³ of the contaminated water
 - (iii) The minimum mass of calcium carbonate needed to neutralize all the Acid in 10,000cm³ of the contaminated water.

Solution

Table of results

Burette reading	Pilot	1	2	3
Final reading (cm ³)	24.90	25.10	24.90	25.00
Initial reading (cm ³)	0.00	0.00	0.00	0.00
Volume of acid used(cm ³)	24.90	25.10	24.90	25.00

$$\text{Average volume of acid used} = \frac{v_1 + v_2 + v_3}{3} = \frac{24.90 + 25.10 + 24.90}{3}$$

$$\text{Average volume} = 25.00 \text{ cm}^3.$$

(a) 25cm³ of the average volume of R required 25cm³ of L for complete neutralization

(b) The colour change at the end point was from Yellow to Pink

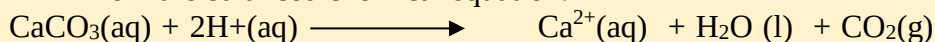
(c) The result of the first titration usually ignored because (*Only one reason*)

i. It is done to show at which range the colour probably change.

ii. it is always inaccurate.

(d) . Concentration of Hydrogen ion in Moles/dm³ = ?

From the balanced chemical equation.



Data given.

Mb = 0.1M

Vb = 25cm³

na = 2

nb = 1

Ma = ?

From

$$\frac{M_a V_a}{M_b V_b} = \frac{n_a}{n_b}$$

$$M_a = \frac{M_b V_b n_a}{n_b V_a}$$

$$M_a = \frac{0.1 \text{ M} \times 25 \text{ cm}^3 \times 2}{1 \times 25 \text{ cm}^3}$$

$$= 0.2 \text{ M}$$

$$= \frac{0.1 \text{ M} \times 25 \text{ cm}^3 \times 2}{1 \times 25 \text{ cm}^3}$$

$$= 0.2 \text{ M}$$

Concentration of Hydrogen ion = 0.2 Moles/dm³

(ii) The number of Moles of hydrogen ion present in 10,000cm³ of contaminated water =?

Since Concentration of hydrogen ion = 0.2Moles/dm³

Therefore

To change cm³ into dm³

$$1\text{dm}^3 = 1000\text{cm}^3$$

$$x = \frac{1 \times 10000}{1000}$$

Since

$$0.2\text{Moles} = 1\text{dm}^3$$

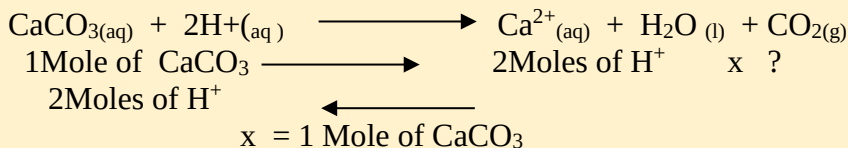
$$? = 10\text{dm}^3$$

$$x = 2\text{Moles}$$

Number of Moles of Hydrogen ion present = 2Moles

(iii) The Minimum mass of Calcium carbonate =?

From the equation;



$$x = 1\text{ Mole of CaCO}_3$$

$$\text{Since Moles} = \frac{\text{Mass given}}{\text{Molar mass}}$$

$$\text{Mass of CaCO}_3 = \text{Moles} \times \text{Molar Mass}$$

$$\begin{aligned} \text{Molar mass of CaCO}_3 &= 40\text{g/mole} + 12\text{g/mole} + (16 \times 3)\text{g/mole} \\ &= \underline{100\text{g/mole}} \end{aligned}$$

$$\begin{aligned} \text{Mass of CaCO}_3 &= 2\text{Moles} \times 100\text{g/mole} \\ &= 200\text{g.} \end{aligned}$$

The Minimum mass of Calcium carbonate = 200g.

PRACTICAL ACTIVITY

1. You are provided with the following:

A: Sodium hydroxide solution, 0.05M.

B: Ethanoic acid solution of unknown concentration

Pop: Phenolphthalein indicator.

Procedure:

- (i) By using **POP** titrate **B** against **A** until the end point is reached.
(ii) record the results in tabular form.

Questions

- (a) (i) State the volume of pipette used.
(ii) State the colour change at the end point
(iii) Find the average volume of B used
- (b) Calculate the number of moles sodium hydroxide used in the titration
- (c) Write an equation for the reaction of sodium hydroxide with ethanoic acid
- (d) (i) Calculate the number of moles of ethanoic acid present in 1dm^3 .
(ii) deduce the number of moles of ethanoic acid present in 25dm^3
(iii) convert the value of moles of ethanoic acid calculated in d(ii) into grams of ethanoic acid.

SOLUTION

- (a) (i) The pipette used was of 25cm^3
(ii) Colour change was from pink to colourless
(iii) Table of results

Burette reading	Pilot	1	2	3
Final reading(cm^3)	25.20	25.00	25.10	25.00
Initial reading(cm^3)	0.00	0.00	0.00	0.00
Volume of acid used(cm^3)	25.20	25.00	25.10	25.00

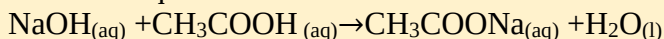
$$\begin{aligned}\text{Average volume} &= \frac{V_1 + V_2 + V_3}{3} \\ &= \frac{25.00 + 25.10 + 25.00}{3} \\ &= 25.00\text{cm}^3\end{aligned}$$

- (b) Number of moles of NaOH
Since 0.05M dissolved in 1dm^3

$$\begin{array}{l} 0.05\text{moles} \quad \underline{\hspace{2cm}} \quad 1\text{dm}^3 \\ ? \quad \underline{\hspace{2cm}} \quad \times \quad \underline{\hspace{2cm}} \quad 0.025\text{dm}^3 \end{array}$$

❖ Number of moles of NaOH was 0.00125 moles

- (c) Balanced equation



- (d) (i) we have

molarity of base(mb)=0.05M

volume of base(vb)=25cm³

volume of acid(ma)=25cm³

number of moles of base(nb)=1

number of moles of acid(na)=1

molarity of acid(ma)=?

From titration formula

$$\frac{m_a v_a}{m_b v_b} = \frac{n_a}{n_b}$$

$$M_a = \frac{m_b v_b}{\frac{v_a}{0.05} \times 25}$$
$$= \frac{0.05 \times 25}{25.00}$$

molarity of acid was 0.05M

number of moles of acid in 1dm³

from molarity = $\frac{n}{v}$

number of moles = M x V = 0.05M x 1dm³

number of moles in 1dm³ = 0.05 moles

(ii) Number of moles of acid in 25cm³

Mole ratio = 1:1

1mole of base → 1mole of acid

0.00125mole → ?

Number of moles of acid = 0.00125 moles

(iii) mass of acid

From the formula

$$\text{Number of moles} = \frac{\text{mass}}{\text{molar mass}}$$

Mass = number of mole X molar mass

= 0.00125mol X 60g/mole

= 0.075g

Mass of an acid was 0.075g

NECTA 2012 CHEMISTRY 2A.

1. You are provided with the following solutions;

TZ: Containing 3.5g of impure sulphuric acid in 500 cm³ of solution;

LO: Contain 4g of sodium hydroxide in 1000 cm³ of solution

Phenolphthalein and methyl orange indicators.

Questions

- a) i. What is the suitable indicator for the titration of the given solution? Give a reason for your answer
ii. write a balanced chemical equation for the reaction between TZ and LO.
iii. Why it is important to swirl or shake the contents of the flask during the addition of acid?
- b) Titrate the acid (in a burette) against the base (in a conical flask) using two drops of your indicator and obtained three titre values.
- c) i. ___ cm^3 of acid required ___ cm^3 of base for complete reaction.
ii. Showing your procedure clearly, calculate the percentage purity of TZ.

SOLUTION

- a) i. The suitable indicator for the titration of the given solution is either of the two indicators, it can be either methyl orange (mo) or phenolphthalein (p. o. p) indicators because the titration involves strong acid against strong base.
ii. $\text{H}_2\text{SO}_4(\text{aq}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{Na}_2\text{SO}_4(\text{aq}) + \text{H}_2\text{O}(\text{l})$
iii. It is important to swirl or shake the contents of the flask during the addition of the acid to allow reaction to take place properly or completely as well as allowing easy recognition of sharp end point during titration.

b) Table of results

Titration number	Pilot	1	2	3
Final burette reading (cm^3)	25.90	25.10	25.40	25.00
Initial burette reading (cm^3)	0.00	0.00	0.00	0.00
Volume used or titre value (cm^3)	25.90	25.10	25.40	25.00

$$\text{Average titre value} = \frac{V_1 + V_2 + V_3}{3}$$

$$V_a = \frac{25.10 + 25.40 + 25.00}{3}$$

$$\text{Average volume} = \underline{25.20 \text{ cm}^3}$$

c. i. 25.20 cm³ of acid solution required 25cm³ of base for complete reaction.

ii. Calculate for the percentage purity of TZ

Given that:

$$\text{Concentration (g/dm}^3\text{)} = 4\text{g/dm}^3$$

$$\text{So, its' Molarity (mol/dm}^3\text{)} = \frac{\text{conc(g/dm}^3\text{)}}{\text{molar mass}(\frac{\text{g}}{\text{mol}})} \\ = \frac{4\text{g/dm}^3}{40\text{g/mol}}$$

Molarity of solution **LO** (base) was 0.1mol/dm³

Molarity of solution **TZ** (mol/dm³) can be obtained from the relation

$$\frac{m_a v_a}{m_b v_b} = \frac{n_a}{n_b}$$

Where:

Ma = Molarity of Acid (TZ)

Va = Volume of Acid (TZ)

Mb = Molarity of Base (LO)

Vb = Volume of Base (LO)

Na = Number of Acid (TZ)

nb = Number of Base (LO)

We have;

$$M_a = ?$$

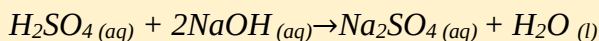
$$V_a = 25.20\text{cm}^3$$

$$M_b = 0.1\text{M}$$

$$V_b = 25\text{cm}^3$$

$$n_b = 2$$

$$n_a = 1$$



$$n_a = 1 \quad n_b = 2$$

$$M_a = \frac{M_b \times V_b \times n_a}{V_a \times n_b}$$

$$M_a = \frac{0.1 \times 25 \times 1}{25.20 \times 2}$$

$$= 0.05\text{M}$$

Molarity of solution **TZ** (acid) = 0.05M.

Then,

Calculate the percentage purity of **TZ**

$$\text{Percentage purity} = \frac{\text{Concentration of pure X} \times 100\%}{\text{Concentration of impure}}$$

Therefore;

Find the concentration of pure in solution TZ

Concentration of pure = Molarity X molar mass

Concentration of pure **TZ** = 0.05 X 98

Concentration of pure **TZ** = 4.9g/dm³

Find the concentration of impure acid in solution TZ

$$\text{Concentration of impure} = \frac{\text{Mass}}{\text{Volume}}$$

$$\text{Concentration of impure of impure} = \frac{3.5g}{0.5dm^3}$$

$$\text{Concentration of impure} = 7g/dm^3$$

Percentage purity = Concentration of pure X 100%

Concentration of impure

$$\text{Percentage purity} = \frac{4.9g/dm^3}{7g/dm^3} \times 100\%$$

$$\text{Percentage purity} = 70\%$$

$$\text{Percentage purity of solution TZ} = 70\%$$

PRACTICAL ACTIVITY

- You are provided with the following solutions;
 Solution **AA** Prepared by diluting 100cm³ of 1M Hydrochloric Acid
 Solution **BB** is sodium hydroxide
 Phenolphthalein indicator (POP)

Procedure:

Pipette 20 cm³ or 25cm³ of solution BB into a titration flask. Add two drops of POP indicator. Titrate solution BB against solution AA from the burette until a colour change is observed. Note the burette reading. Repeat the procedure to obtain three more readings. Record your results as shown in the table below.

- (i) Table of results.

Titration number	Pilot	1	2	3
Final reading (cm ³)				
Initial reading (cm ³)				

Volume used (cm ³)				
--------------------------------	--	--	--	--

- ii) The volume of pipette used was ____ Cm³
 iii) The colour change at the end point was from ____ to ____
 iv) ____ Cm³ of solution **AA** was required to neutralize ____ Cm³ of solution **BB**
 b) Write a balanced chemical equation for the neutralization of sodium hydroxide by hydrochloric acid
 c) Calculate the:
 (i) Molarity of solution **AA**
 (ii) Concentration in mole/dm³ of solution **BB**
 (iii) Concentration in g/dm³ of solution **BB**

SOLUTION

- a) Table of results
 i) Table of results

Burette reading	Pilot	1	2	3
Final reading (cm ³)	24.00	48.10	23.70	47.00
Initial reading (cm ³)	0.00	24.00	0.00	23.70
Volume used (cm ³)	24.00	24.10	23.70	23.30

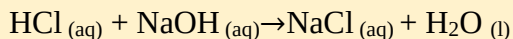
$$\text{Average volume of acid (AA)} = \frac{V_1 + V_2 + V_3}{3}$$

$$V_a = \frac{24.10 + 23.70 + 23.30}{3}$$

$$V_a = \underline{23.70 \text{ cm}^3}$$

- ii) The volume of pipette used was 25Cm³
 v) The colour change at the end point was from Pink to Colourless
 vi) 25 Cm³ of solution **AA** was required to neutralize 23.70 Cm³ of solution **BB**

- b) Balanced chemical equation



c)

(i) Molarity of solution AA

From the dilution formula

$$M_c V_c = M_d V_d$$

Where

M_c = Molarity of conc. Acid (AA)

V_c = Volume of conc. Acid (AA)

M_d = Molarity of diluted Acid (AA)

V_d = Volume of diluted or final volume of Acid AA

We have;

$$M_c = 1M$$

$$V_c = 100\text{cm}^3$$

$$V_d = 1000\text{cm}^3$$

$$M_d = ?$$

$$M_c V_c = M_d V_d$$

$$1M \times 100\text{cm}^3 = M_d \times 1000\text{cm}^3$$

$$M_d = \frac{1M \times 100\text{cm}^3}{1000\text{cm}^3}$$

$$= 0.1M$$

❖ **Molarity of dilute solution AA was 0.1M**

(ii) Concentration in mole/dm³ of solution BB

From the relation

$$\frac{M_a V_a}{M_b V_b} = \frac{n_a}{n_b}$$

Where:

M_a = Molarity of Acid (AA)

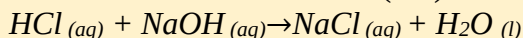
V_a = Volume of Acid (AA)

M_b = Molarity of Base (BB)

V_b = Volume of Base (BB)

n_a = Number of Acid (AA)

n_b = Number of Base (BB)



$$n_a = 1 \quad n_b = 1$$

We have

$$M_a = 0.1M$$

$$V_a = 23.70\text{cm}^3$$

$$M_b = ?$$

$$V_b = 25 \text{ cm}^3$$

$$n_b = 1$$

$$n_a = 1$$

$$M_b = \frac{M_a \times V_a \times n_b}{V_b \times n_a}$$

$$M_b = \frac{0.1 \times 23.70 \times 1}{25 \times 1}$$

$$M_b = 0.0948 \text{ M} = 0.1 \text{ M}$$

Concentration of BB in mol/dm³ was 0.1M.

(iii) Concentration in g/dm³ of solution BB

From the relation

$$\text{Molarity (mol/dm}^3\text{)} = \frac{\text{Conc in g/dm}^3}{\text{Molar mass g/mol}}$$

$$\text{Molar mass of solution BB} = 40 \text{ g/mol}$$

$$\text{Concentration in g/dm}^3 = \text{Molarity} \times \text{molar mass}$$

$$\text{Concentration in g/dm}^3 = 0.1 \times 40$$

$$\text{Concentration in g/dm}^3 = 4 \text{ g/dm}^3$$

❖ **Concentration in g/dm³ of solution BB was 4 g/dm³**

PRACTICAL PROBLEM

1. You are provided with the following solutions;

Solution E: Containing 3.65g of Hydrochloric Acid per dm³ of solution.

Solution K: Containing 3.575g of pure hydrated sodium carbonate, Na₂CO₃.xH₂O per 0.25dm³ of solution.

Methyl orange indicator solution

Procedure:

Put the acid solution into the burette. Pipette 20 cm³ or 25cm³ of solution **K** into a titration flask. Add few drops of methyl orange indicator. Titrate this base solution against the acid solution until an end point is reached. Record your results as shown below and Repeat the titration.

- a) Table of results

Titration number	Pilot	1	2	3
Final reading (cm ³)				

Initial reading (cm ³)				
Volume used (cm ³)				

The volume of pipette used was ____ Cm³

- i. The colour changed at the end point was from ____ to ____
- i. The volume of acid solution E needed for complete neutralization was ____ Cm³
 - ii. Write down the balanced chemical equation for the reaction.
- b) Calculate the:
- i. Molarity of solution E
 - ii. Molarity of solution K
 - iii. Find the value of water crystallization in Na₂CO₃. XH₂O

SOLUTION

The volume of pipette used was 20 Cm³

a) Table of results

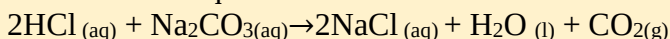
Titration number	Pilot	1	2	3
Final reading (cm ³)	20.00	39.80	19.60	40.00
Initial reading (cm ³)	0.00	20.00	0.00	20.00
Volume used (cm ³)	20.00	19.80	19.60	20.00

$$\text{Average volume of acid (E)} = \frac{V_1 + V_2 + V_3}{3}$$

$$V_a = \frac{19.80 + 19.60 + 20.00}{3}$$

$$V_a = \underline{19.90 \text{ cm}^3}$$

- b) i. The colour change at the end point was from Yellow to pink
- ii. Balanced equation for the reaction



- C) Calculate the;
- (i) Molarity of Acid Solution E

From the relation;

$$\text{Molarity (mol/dm}^3\text{)} = \frac{\text{Concing/dm}^3}{\text{Molarmassg/mol}}$$

Then;

Calculate the concentration of solution E

$$\frac{mava}{mbvb} = \frac{na}{nb}$$

Where:

Ma = Molarity of Acid (FF)

Va = Volume of Acid (FF)

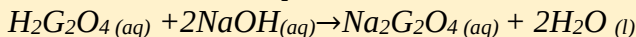
Mb = Molarity of Base (EE)

Vb = Volume of Base (EE)

na = Number of moles of Acid (FF)

nb = Number of moles of Base (EE)

Balanced chemical equation



$$n_a = 1 \quad n_b = 2$$

We have

$$M_a = ?$$

$$V_a = 25.00 \text{ cm}^3$$

$$M_b = 0.4M$$

$$V_b = 25 \text{ cm}^3$$

$$n_b = 2$$

$$n_a = 1$$

$$M_a = \frac{M_b \times V_b \times n_a}{V_a \times n_b}$$

$$M_a = \frac{0.4 \times 25.00 \times 1}{25.00 \times 2}$$

$$M_a = 0.2M$$

Molarity an acid was 0.2M.

Step 2:

Find the molar mass of $H_2G_2O_4$

$$\text{Molar mass (g/mol)} = \frac{\text{Conc}(\frac{g}{dm^3})}{\text{Molarity (mol/dm}^3\text{)}}$$

$$\text{But concentration of } H_2G_2O_4 = \frac{\text{mass}}{\text{volume}}$$

$$= \frac{4.50g}{0.25dm^3} = 18g/dm^3$$

$$\text{Molar mass } H_2G_2O_4 = \frac{18g/dm^3}{0.2mol/dm^3}$$

$$\text{Molar mass of } H_2G_2O_4 = 90g/mol$$

Then;

$$H_2G_2O_4 = 90$$

$$2 + 2G + (16 \times 4) = 90$$

$$66 + 2G = 90$$

$$2G = 90 - 66$$

$$G = 24/2$$

$$G = 12$$

Therefore; Element with atomic mass 12 is carbon

Hence, element G was carbon (acid was oxalic acid)

NECTA 2015 CHEMISTRY 2A

1. You are provided with the following solutions;
G: Containing 0.1 mole hydrochloric acid per dm^3 of solution;
B: Contain 2.65g M_2CO_3 per 0.5dm^3 of solution
Methyl orange indicator.

Questions;

- (a) Titrate G (in burette) against B (in conical flask) using two drops of your indicator obtain three titre values. Record your data in a tabular form.
- (b) (i) ____ cm^3 of B required ____ cm^3 of G for complete reaction.
(ii) Write a balanced chemical equation for the reaction between **B** and **G** and the corresponding ionic equation with state symbols.
(iii) Showing your procedures clearly; calculate the molar mass of M_2CO_3 and hence identify element M.

SOLUTION

(a) Table of results

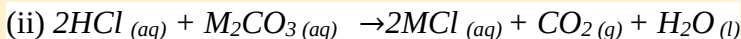
Burette reading	Pilot	1	2	3
Final reading (cm^3)	26.00	25.00	25.10	25.00
Initial reading (cm^3)	0.00	0.00	0.00	0.00
Volume used (cm^3)	26.00	25.00	25.10	25.00

$$\text{Average volume of acid (E)} = \frac{V_1 + V_2 + V_3}{3}$$

$$V_a = \frac{25.00 + 25.10 + 25.00}{3}$$

$$V_a = 25.00 \text{ cm}^3$$

- (b) (i) 25cm^3 of B required 25.00cm^3 of G for complete reaction



(iii) Calculate the molar mass of M_2CO_3 and hence identify element **M**.

From the formula;

$$\frac{MaVa}{mbvb} = \frac{na}{nb}$$

Where:

Ma = Molarity of Acid

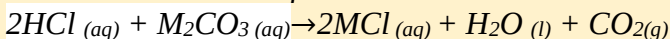
Va = Volume of Acid

Mb = Molarity of Base

Vb = Volume of Base

na = Number of Acid

nb = Number of Base



$$n_a = 2 \quad n_b = 1$$

We have

$$Ma = 0.1M$$

$$Va = 25.03 \text{ cm}^3$$

$$Mb = ?$$

$$Vb = 25.00 \text{ cm}^3$$

$$nb = 1$$

$$na = 2$$

$$\text{from} \quad M_b = \frac{MaVanb}{Vbna}$$

$$M_b = \frac{0.1 \times 25.00 \times 1}{25.00 \times 2}$$

$$M_b = 0.05 \text{ M}$$

Molarity of base (M_b) = 0.05M.

Then;

$$\text{Concentration of } M_2CO_3 = \frac{\text{Mass}}{\text{Volume}}$$

$$\text{Concentration of } M_2CO_3 = \frac{2.65g}{0.5dm^3}$$

$$\text{Concentration of } M_2CO_3 = 5.3g/dm^3$$

But

T=56-17

T=39

Molar mass of element T was 39g/mol

Element T with molar mass of 39 is potassium

NECTA 2011

2. You are provided with the following;

AA: Containing 0.2 mole HNO_3 per dm^3 of solution;

BB: Contain 4.2g Na_2CO_3 per 0.5dm^3 of solution

MO is Methyl orange indicator.

Procedure:

Put solution **AA** into the burette. Pipette 20 cm^3 or 25cm^3 of solution **BB** into a titration flask. Add two drops of methyl orange indicator into the titration flask. Titrate solution **BB** against solution **AA** from the burette until a colour change is observed. Note the burette reading. Repeat the procedure to obtain three more readings and Record your results as in a tabular form.

Questions:

- (a) (i) Calculate the average titre volume.
(ii) Summary 25cm^3 of solution **BB** required 12.50 cm^3 of solution **AA** for complete reaction.
- (b) If the mole ratio for the reaction is 1:1 find:
(i) Concentration of Na_2CO_3 in mol/dm^3 and g/dm^3 .
(ii) Molecular mass of Na_2CO_3 .
(iii) Atomic mass of x and replace it in the formula Na_xCO_3 .
- (c) Write a balanced chemical equation for the reaction in this experiment.
- (d) What is the significance of the indicator in this experiment?
- (e) Why is there a colour change when enough acid has been added to the base?

NECTA 2014

1. You are provided with the following solutions;

H: Containing 6.3 g of hydrated oxalic acid,

$(\text{COOH})_2 \cdot x\text{H}_2\text{O}$ in 1 dm^3 of solution.

M: Contain 1.4 g of potassium hydroxide in 0.5dm^3 of solution.

CHAPTER 2.

CHEMICAL KINETICS AND EQUILIBRIUM

Chemical kinetics-is the study of the rate of chemical reactions.

Chemical equilibrium –refers to the relative stability of the reactants and the products.

Rate of Chemical Reaction is the measure of the progress of the reaction per unit time. It determined by measuring number of reactants used or products formed.

A slow reaction is the one which has low rate of reaction. It takes a much longer time to complete the reaction.

A rapid reaction is the one which has higher rate of reaction. It takes a short time for the reaction to be completed.

$$\text{Rate} = \frac{1}{\text{time}} (\text{sec}^{-1})$$

Factors affecting the rate of chemical reaction.

The following are factors that affect the rate of chemical reaction.

- ❖ Change in Temperature
- ❖ change in Concentration
- ❖ Surface area of reactant
- ❖ Addition of Catalyst
- ❖ Change in Pressure

Only the first two factors will be discussed at this level.

Effect of Concentration of Reactants on The Rate of Chemical Reaction.

The rate of a reaction is proportional to the concentration of reactants. It is also qualitatively agreed that for a chemical reaction to occur, reacting particles (molecules) must first come into contact or collide.

Based on these two arguments, it can be concluded that, there is a greater chance that molecules will collide more in a system where particles (molecules) are in higher concentration than in the system where they are less concentrated.

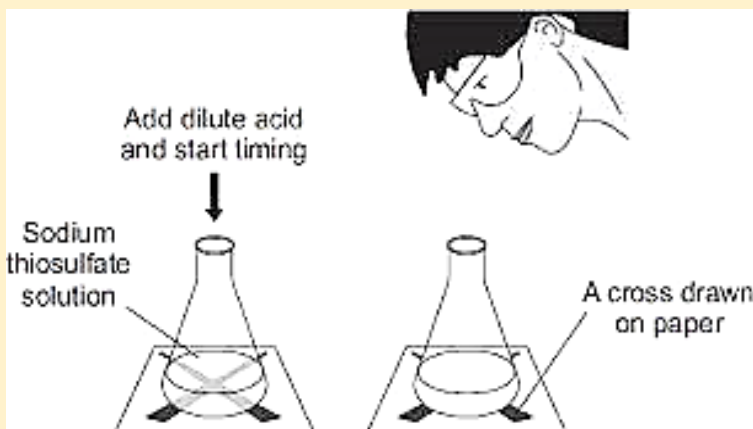
Chemicals used in this practical

- ❖ Sodium thiosulphate ($\text{Na}_2\text{S}_2\text{O}_3$)
- ❖ Hydrochloric acid (HCl)
- ❖ Nitric acid (HNO_3)

Note

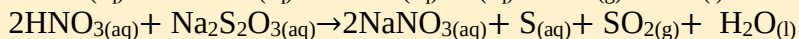
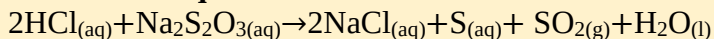
A chemical reaction will occur only if the particles of reacting substances are allowed to come into contact.

When nitric acid or hydrochloric acid reacts with sodium thiosulphate, precipitates of sulphur will be formed which cause cross to obscure (cloud)



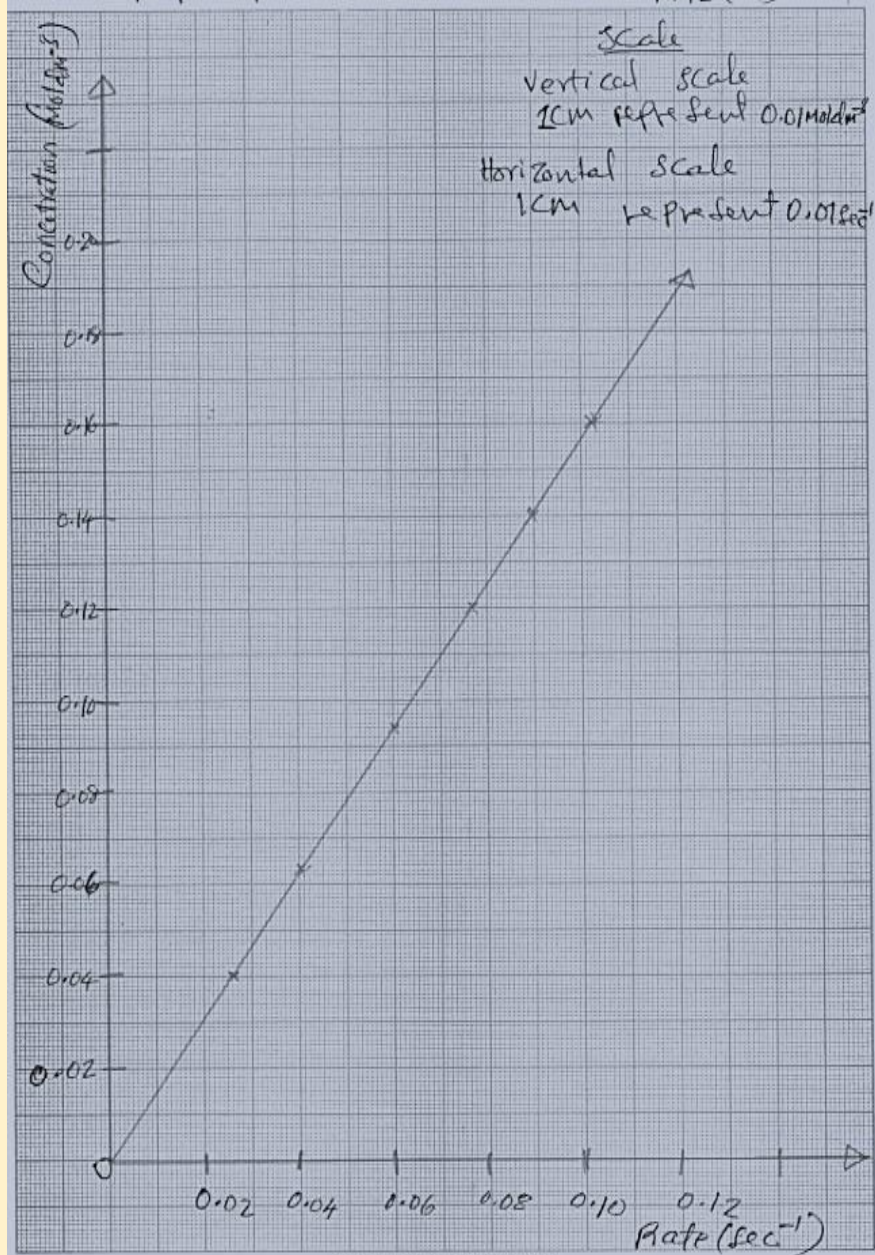
The fast reaction (higher rate of chemical reaction) takes short time for the precipitates of Sulphur to be formed and vice versa

Balanced chemical equations

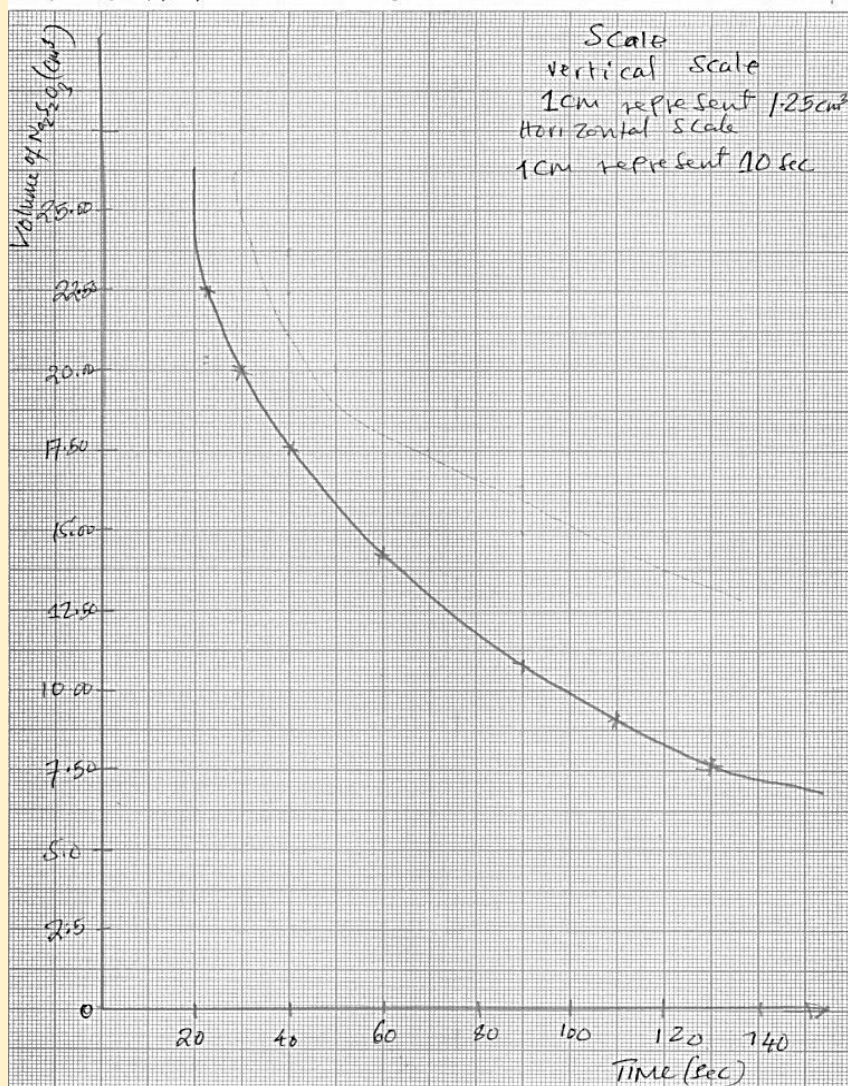


NATURE OF GRAPHS

A GRAPH OF CONCENTRATION AGAINST RATE (sec^{-1})



A GRAPH VOLUME (CM³) AGAINST TIME (SEC)



Effect of Temperature on The Rate Chemical Reaction

By increasing the temperature of a reaction, the heat supplied to the reacting particles, increases the kinetic energy. This increases the frequency of collisions of particles. Increasing kinetic energy of particles hence speeds up the rate of reaction as it provides the activation energy.

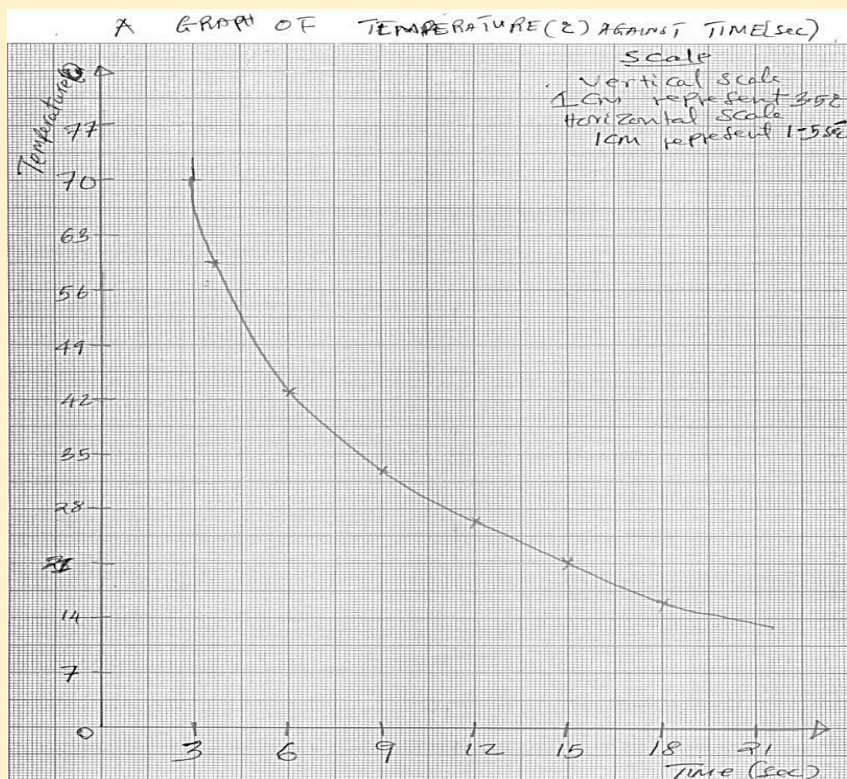
Therefore, increase in temperature increases the rate of chemical reaction and vice versa.

At higher temperature, the time for disappearance of cross(X) decreases. The cross X disappears due to formation of fine particles of sulphur.

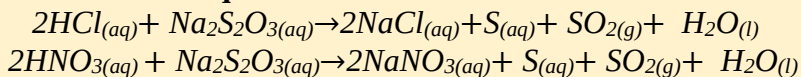
Chemicals used.

- ❖ Sodium thiosulphate ($\text{Na}_2\text{S}_2\text{O}_3$)
- ❖ Hydrochloric acid (HCl)
- ❖ Nitric acid (HNO_3)

Nature Of Graphs



Balanced chemical equations.



Uses of Sulphur

- ❖ Most of the Sulphur produced in the world is used to make sulphuric acid.
- ❖ Sulphur is used in the manufacture of Sulphur dioxide and sodium sulphite, which are used for bleaching straw and wood fibres. They are also used in removing lignin from wood pulp in the paper industry.
- ❖ Gunpowder is a mixture of potassium nitrate, carbon and Sulphur.
- ❖ Sulphur is used in the vulcanization of natural rubber. Vulcanization is the process of making naturally soft rubber harder by reacting it with Sulphur.
- ❖ Sulphur is used in the manufacture of various organic compounds such as plastics and medicines.

NOTE:

Atomic number of Sulphur is 16

Electronic configuration of Sulphur is 2:8:6

NECTA 2014 CHEMISTRY 2B

2. You are provided with the following;

X: containing 20g of solid sodium thiosulphate per litre

Y: 2.0 mol dm⁻³ hydrochloric acid;

Distilled water;

Stop-watch;

Thermometer;

Plain paper.

Procedure:

- Draw a cross on white paper.
- Measure 20cm³ of solution **x** into 100cm³ beaker. Add 20cm³ of distilled water and heat the contents with a gentle flame.

- iii. When the temperature reaches 60°C , transfer the beaker on top of a cross and immediately add 5cm^3 of solution **Y**. at the same time start the stop watch.
- iv. Stir the mixture with glass rod, while observing the cross on top view
 - i. Stop the clock immediately as soon as the cross is obscured and record the time.
 - ii. Repeat the experiment three times at a temperature of 50°C , 40°C and room temperature as indicated in table 1

Table 1

Experiment number	Temperature ($^{\circ}\text{C}$)	Concentration of sodium thiosulphate (mol dm^{-3})	Time, t (sec)
1	60	0.0635	
2	50	0.0635	
3	40	0.0635	
4	Room temperature	0.0635	

Questions

- (a) What is the aim of this experiment?
- (b) Complete the table 1.
- (c) Giving reason(s), identify the experiment in which the reaction was;
 - (i) fast
 - (ii) slow
- (d) with state symbols, write the ionic equation for the reaction between **X** and **Y**
- (e) (i) draw the graph of temperature against time
 (ii) what conclusion can you draw from the graph?
 Give reason for your answer.

SOLUTION

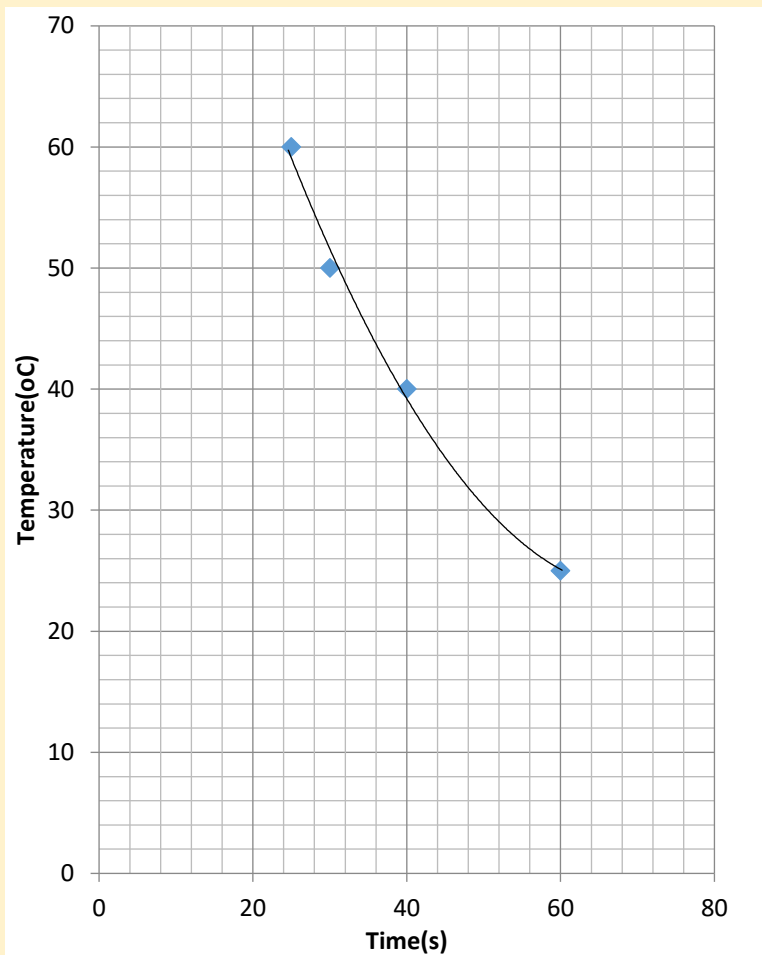
- (a) Aim of the experiment was to investigate the effect of temperature on the rate of chemical reaction.
- (b) Table of results.
Table 1

Experiment number	Temperature (°C)	Concentration of sodium thiosulphate (mol dm ⁻³)	Time, t (sec)
1	60	0.0635	25
2	50	0.0635	30
3	40	0.0635	40
4	Room temperature 25	0.0635	60

- (c) (i) experiment number 1 was fast due to high temperature which increase collision frequency of reactant particles.
(ii) Experiment which number 4 slow due to low temperature and low collision frequency.
- (d) Balanced chemical equation

$$2\text{HCl}_{(\text{aq})} + \text{Na}_2\text{S}_2\text{O}_{4(\text{aq})} \rightarrow 2\text{NaCl}_{(\text{aq})} + \text{S}_{(\text{s})} + \text{SO}_{2(\text{g})} + \text{H}_2\text{O}_{(\text{aq})}$$
Ionic equation

$$2\text{H}^+_{(\text{aq})} + \text{S}_2\text{O}_3^{2-}_{(\text{aq})} \rightarrow \text{S}_{(\text{s})} + \text{SO}_{2(\text{g})} + \text{H}_2\text{O}_{(\text{aq})}$$
- (e) (i) graph of temperature(°C) against time(S)



(ii) As temperature decreases the time taken for disappearance of cross increases.

**NECTA 2012
CHEMISTRY 2B**

2. You are provided with the following:

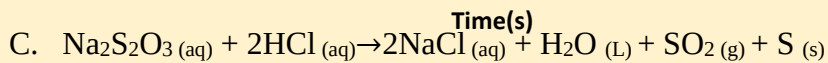
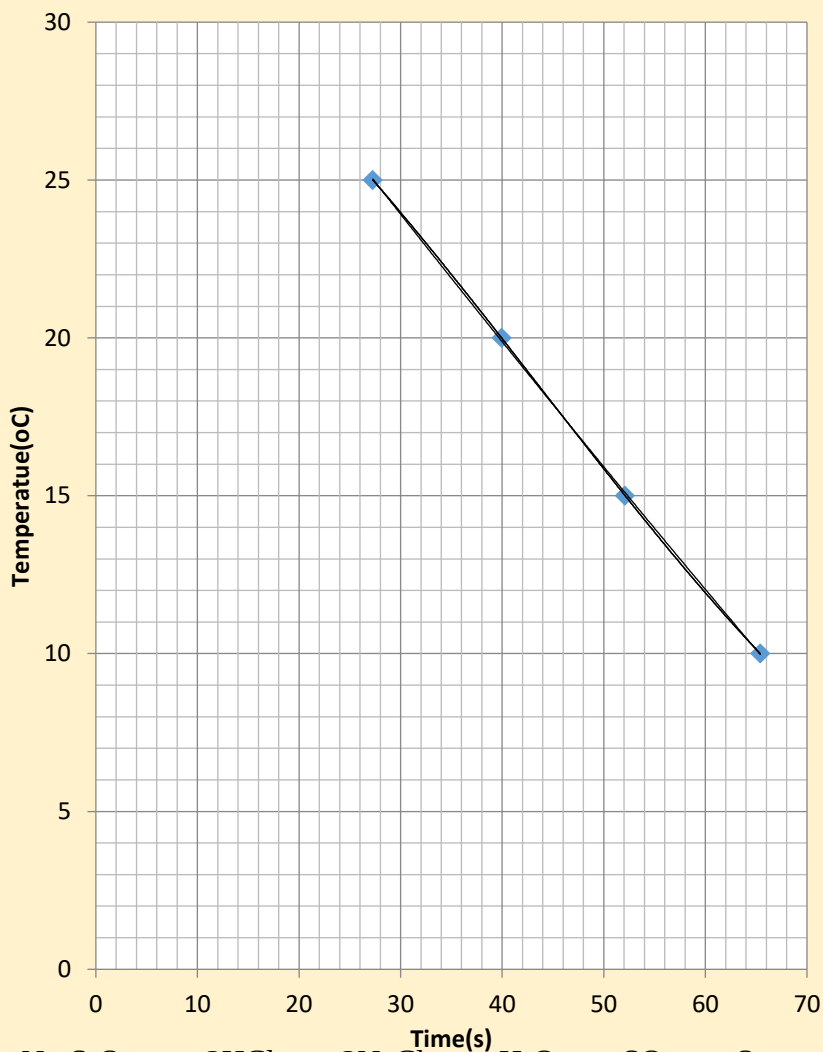
ZO: A solution of 0.13M of $\text{Na}_2\text{S}_2\text{O}_3$

UU: A solution of 2M HCl;

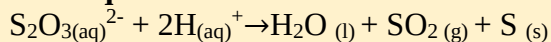
Thermometer

Heat source/ burner

Stop watch



In ionic equation



d. 1/time represent rate of chemical reaction.

- e. As temperature of the reactants increases, the rate of chemical reaction is increases or temperature is directly proportional to the rate of chemical reaction.

NECTA 2017

2. You are provided with the following:

E: A solution containing 79g of $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$

F: A solution contain 0.1mol/dm^{-3} of dilute HCL

G: Distilled water

A stop watch/ clock; a white plain paper and thermometer.

Procedure

- (i) Put a beaker (100cm^3) on top of a cross on the given sheet of paper.
 - (ii) Measure 25 cm^3 of E using measuring cylinder and pour into a beaker in (i).
 - (iii) Using another measuring cylinder measure 5 cm^3 of F and pour it into a beaker containing E and instantly, start a stop watch.
 - (iv) Stir the mixture with a glass rod while you keep on observing the cross from above; record the time taken for the cross to disappear.
 - (v) Repeat the procedures for different concentrations of E by taking 20 cm^3 , 15 cm^3 , 10 cm^3 and 5 cm^3 of the original E and making the total volume up to 25 cm^3 by adding G.
- Repeat the results as shown in the table 1.

Table 1

Volume of E (cm^3)	Volume of G (cm^3)	Time taken for the cross to Disappear (sec)
25		
20		
15		
10		
5		

Questions

- a) Complete filling the Table 1

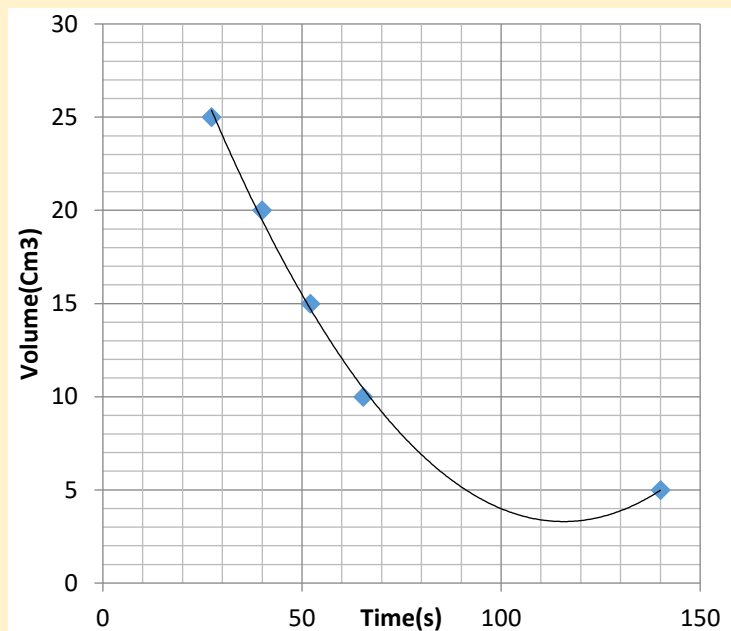
- b) (i) Using the data in the table, plot a graph of volume against time graph (volume on the y-axis and time in second on the x-axis)
 (ii) What does the shape of the graph indicate?
 c) Write down the ionic equation of the reaction between E and F.
 d) Why did the cross disappear?
 e) Write two uses of the product which obscured the cross.

SOLUTION

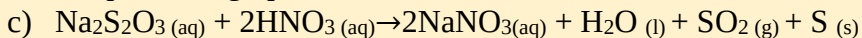
a) Table of results

Volume of E (cm ³)	Volume of G (cm ³)	Time taken for the cross to disappear (sec)
25	0	27.27
20	5	39.94
15	10	52.10
10	15	65.40
5	20	140.00

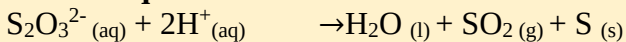
b) (i)



(ii) The shape of the graph is a curve



In ionic equation



d) Cross disappear because of Precipitation fine particles of sulphur

- ❖ used to make sulphuric acid
- ❖ used in the manufacture of gunpowder
- ❖ used in the manufacture of sulphur dioxide which is used for bleaching straw and wood fibres
- ❖ used in vulcanization of natural rubber
- ❖ used in the manufacture of various organic compounds such as plastics and medicines
- ❖ used to dust vines to prevent the growth of fungus

NECTA 2018 CHEMISTRY 2A

2. You are provided with the following:

H₁: A solution of 0.25 mol/dm⁻³ Na₂S₂O₃;

H₂: A solution of 2M HCL;

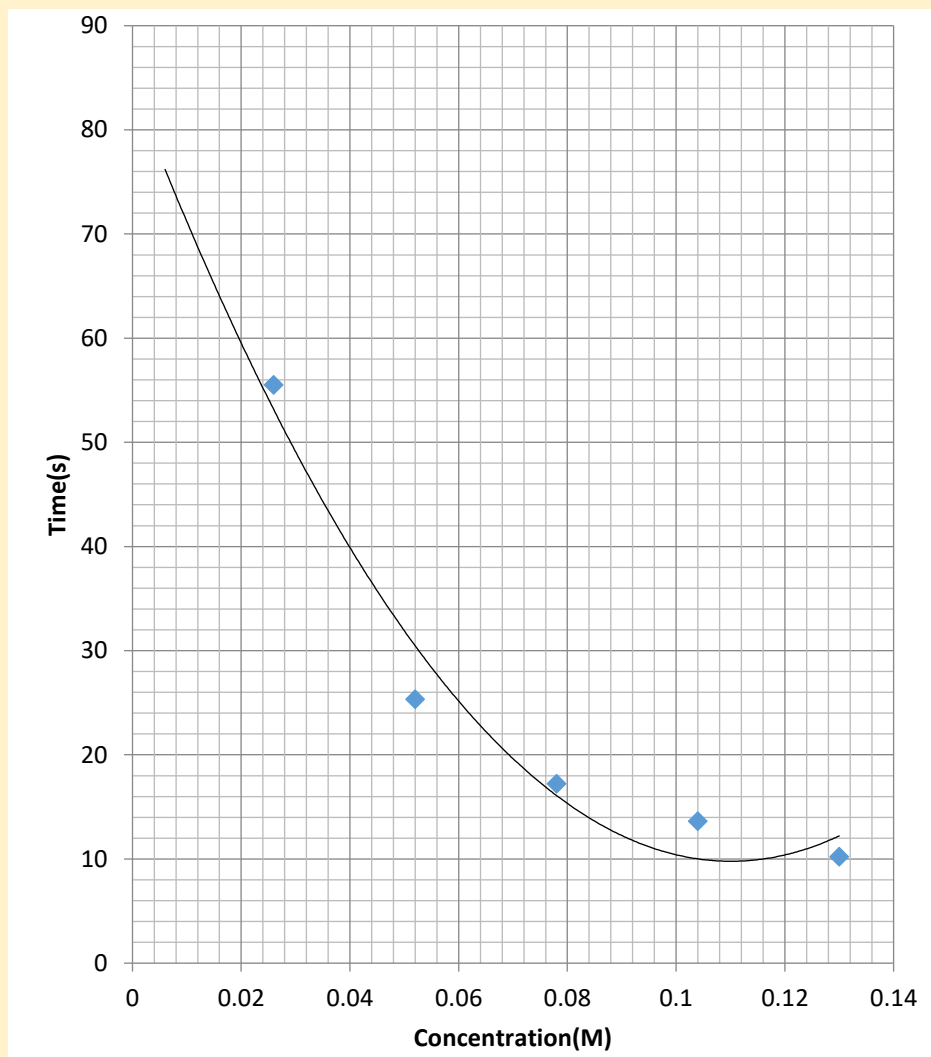
Distilled water, stop watch, Thermometer and a plain paper marked

Procedure.

Place 150cm³ of water in the 250cm³ beaker and use this as your water bath; heat the water 80°C. Measure 10cm³ of H₁ and 25cm³ of water and pour the contents into the 100cm³ beaker. Put the beaker with the contents into a hot water bath. When the contents attain a temperature of 70°C, place the beaker on top of the marker X on the paper provided. Add 10cm³ of H₂ and immediately start the stop watch. Swirl the beaker twice and look vertically on the top of the beaker so as to see X through the bottom of the beaker. Stop the clock when X is invisible. Record the time taken for X to disappear completely. Repeat the experiment at different temperatures as shown in the table 1.

Table 1

Exp. No.	Temperature of H ₁ (°C)	Time (sec)	Rate 1/ t (S ⁻¹)
1	70		
2	60		



d) Substance that was produced during the reaction which obscured the mark A was fine particles of Sulphur (S) (precipitation of sulphur)

e) Conclusion; As concentration of sodium thiosulphate decreases the time taken for disappearance of letter A increases.

f) The smell was due to production of Sulphur (S)

g) Increase in concentration of sodium thiosulphate increases the rate of chemical reaction

Chapter 3

QUALITATIVE ANALYSIS.

Qualitative analysis is a method used for identification of ions or compounds in a sample. It aims at determining substances present in a mixture but not their quantities. By conducting appropriate tests and applying logic, the identities of ions present in an unknown solution can be determined. The analyses to be performed are based upon the idea that no two ions produce the same set of chemical reactions as each ion reacts on its own characteristic way.

In qualitative analysis several types of chemical reactions have to be monitored which are revealed through;

- ❖ Colour change,
- ❖ Evolution of a gas,
- ❖ Production of sound,
- ❖ Increase in temperature of the reaction vessel and
- ❖ The formation of the precipitate (a solid product).

In many cases, qualitative analysis will also involve the separation of ions or compounds in a mixture. Examples of qualitative tests would include ion precipitation reactions (solubility tests) and chemical reactivity tests.

Preliminary tests

There are separate procedures for detecting cations and anions called the cation and anion analysis, respectively. In order to identify the ions, present in the sample, the first thing to perform is preliminary tests. These tests may consist of;

- ❖ Physical examination (colour, texture and smell),
- ❖ dry heating the sample to detect the presence of some constituents such as carbon (marked by the appearance of smoke or char) or water (marked by the appearance of moisture).
- ❖ It may also involve introducing the sample into a flame and noting the colour produced as certain elements may be identified by means of their characteristic flame colours. Other preliminary tests include the action of dilute and concentrated acids as well as solubility in water.

The analytical procedures, experiments/ tests performed have to be recorded in tabular form with three columns titled experiment, observation and inference. The experimental column summarises

the test performed to the unknown sample. This has to be reported in a simple past tense. For example, “a small amount of the sample was picked with a clean spatula to the test tube where a small amount of water was added.” The changes observed when the sample is tested have to be written in the observation column. Examples of the observations include

- ❖ The colour change,
- ❖ Evolution of gases,
- ❖ Smell,
- ❖ Flame colour,
- ❖ Sound and
- ❖ Formation of precipitates.

The suggestions from the observation is written in the inference column. Checking the physical characteristics of a salt, such as colour changes.

Importance of qualitative analysis.

- ❖ Used in finding poisonous substances in the environment.
- ❖ Used to find out ions present in a chemical substance.
- ❖ Used in identification of the pH of a soil sample.
- ❖ Performing medical examinations such as blood and urine tests
- ❖ Identifying gases evolved during chemical reaction

Types of qualitative analysis.

There are two types of qualitative analysis, which are;

- ❖ Guided analysis
- ❖ Open analysis

Guided analysis.in this type of analysis a student is given some instructions that help him/her to identify ions in a given salt sample.

Open analysis. .in this type of analysis, a student is supposed to go through all systematic qualitative analysis procedures to reach conclusion.

Physical properties of salts.

(a)**Physical state**–used to identify the state and texture of the substance.

Ionic salts usually have a crystal shape while sulphate and metal oxide are usually in powder form.

(b) **Colour of the salt**–Used to identify the cation, anion or compound.

Salt	Colour in solid state	Colour in aqueous state
Sodium, Calcium, Zinc, lead (II), Ammonium.	Colourless	colourless
Copper (II)sulphate,Copper (II) nitrate	Blue	Blue
Copper (II) Chloride	Green	Green
Iron (II)sulphate, Iron (II)Nitrate, Iron (II) Chloride	Light green	Light –green
Iron(III) sulphate, Iron(III)Nitrate, Iron(III)Chloride	Brown	Brown

(c) Solubility of salt in water.

Compound	Solubility of salt in water.
Sodium, and Ammonium.	All are soluble
Nitrate salts	All are soluble
Chloride salts	All are soluble except lead (II) chloride which are soluble in hot water.
Sulphate salts.	All are soluble except calcium sulphate, lead (II) sulphate.
Carbonates	All are insoluble except sodium carbonate and ammonium carbonate.

Effect of heat on a salt.

Ammonium salts, carbonates, nitrates and some sulphates decompose on heating. They may exhibit a change in colour or release gases.

When a salt is heated, the released gases gives information about the type of anion the salt contains except for ammonium.

The change in colour of the residue after heating gives the information about the type of cation the salt contains.

Salt	Effect on heat.
------	-----------------

Carbonate salts	
Sodium carbonate	Not decomposed by heat
Calcium Carbonate Zinc Carbonate Lead (II) Carbonate Copper (II) Carbonate	Decompose to give metal oxide and carbon dioxide gas $\text{CaCO}_{3(s)} \rightarrow \text{CaO}_{(s)} + \text{CO}_{2(g)}$ $\text{PbCO}_{3(s)} \rightarrow \text{PbO}_{(s)} + \text{CO}_{2(g)}$
Nitrate salts	
Zinc Nitrate Lead (II) Nitrate Copper (II) Nitrate Calcium Nitrate	$2\text{Zn}(\text{NO}_3)_{2(s)} \rightarrow 2\text{ZnO}_{(s)} + 4\text{NO}_{2(g)} + \text{O}_{2(g)}$ $2\text{Pb}(\text{NO}_3)_{2(s)} \rightarrow 2\text{PbO}_{(s)} + 4\text{NO}_{2(g)} + \text{O}_{2(g)}$ $2\text{Cu}(\text{NO}_3)_{2(s)} \rightarrow 2\text{CuO}_{(s)} + 4\text{NO}_{2(g)} + \text{O}_{2(g)}$ $2\text{Ca}(\text{NO}_3)_{2(s)} \rightarrow 2\text{CaO}_{(s)} + 4\text{NO}_{2(g)} + \text{O}_{2(g)}$

Flame tests

Cations produce characteristic colours of flame. The colour of the flame can be used to identify the cations present. nichrome wire or glass rod is dipped in conc. HCl to allow a small salt sample to adhere on it and heated on a flame.

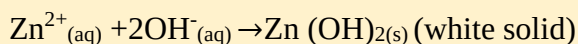
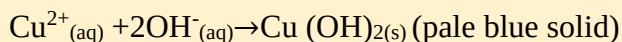
Ion	Colour of the flame.
Na^+	Golden yellow.
Ca^{2+}	Brick red .
Cu^{2+}	Bluish-green.
Pb^{2+}	Blue-white(pale-blue).
Fe^{2+} and Fe^{3+}	Yellow(orange)sparks.
Zn^{2+} and NH_4^+	No definite colour.

Stock solution is a large volume of common reagents such as NaOH or NH_4OH at a standardized concentration from which one can dilute into a working concentration.

(i) Test with aqueous sodium hydroxide

Dilute sodium hydroxide is added drop-wise until in excess and observation made. The substance to be analyzed should be in solution form. Sodium hydroxide contains hydroxide hydroxyl ion (OH^-) which react with metal ions to form insoluble metal hydroxides.

Some of reactions leading to formation of the insoluble metal hydroxides are represented as follows.



(ii) Test with aqueous ammonia.

Dilute ammonium hydroxide is added drop-wise until in excess and observation made. The substance to be analysed should be in solution form. ammonium hydroxide contains hydroxide hydroxyl ion (OH^{-}) which react with metal ions to form insoluble metal hydroxides of different colours.

Cations and anions tested at this level include;

CATIONS	ANIONS
NH_4^{+}	NO_3^{-}
Na^{+}	SO_4^{2-}
Ca^{2+}	Cl^{-}
Pb^{2+}	CO_3^{2-}
Fe^{2+} and Fe^{3+}	HCO_3^{-}
Zn^{2+}	
Cu^{2+}	

The candidates are required to record experiments in tabular form. Experiments should be reported in simple past tense (in most cases in passive voice)

Format of recording analytical experiments.

Experiment	Observation	Inference

Observations are changes which can be seen when a compound is subjected to an experiment or test.

Inference. The tests and observation can help to make logical inferences about the presence or absence of certain in a given salt sample.

QUALITATIVE ANALYSIS PROCEDURES.

(a) **Preliminary tests** are generally for solid samples. The tests include;

- Appearance (colour, texture and deliquescence),
- Flame test,
- Action of heat,
- Solubility in water,
- Action of dilute and
- Action of concentrated acids.

(b) **Tests in solution.**

The tests include;

- Action of dilute NaOH solution
- Action of NH₃ solution.

(c) **Confirmatory tests.**

The preliminary tests and the detection of ions help you to detect the presence of anions and cations present in a salt sample.

The ions to be confirmed is that which has reappeared for several times.

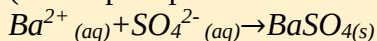
CONFIRMATORY TEST OF ANIONS

1.sulphate radical (SO₄²⁻)

Barium chloride solution (BaCl₂)

Dilute hydrochloric acid

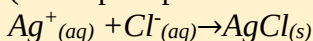
(White precipitate of Barium sulphate is formed)



2.Chloride ion (Cl⁻)

Silver nitrate (AgNO₃) solution

(white precipitate of silver chloride is formed)



3.Nitrate radical (NO₃⁻)

Iron II sulphate (FeSO₄) solution

Conc. Sulphuric acid

(Brown ring is formed)

4.Carbonate (CO₃²⁻) and hydrogen Carbonate (HCO₃⁻) radical

Magnesium sulphate solution (MgSO_4)

(white precipitate is formed before warming which confirms CO_3^{2-}) (white precipitate is formed after warming which confirms HCO_3^-)

Or Dilute nitric acid (HNO_3) (Effervescences of a colourless gas evolves)

or

Barium Chloride solution (BaCl_2) followed by addition of dilute HCl (white precipitate soluble in dilute HCl is formed)

CONFIRMATORY TEST OF CATIONS

Cation	Reagent(S)
Ca^{2+}	(i)Ammonia solutions (NH_4OH) followed by addition of ammonium oxalate. (white precipitate) (ii)Flame test (brick red flame)
Pb^{2+}	(i)Potassium chromate solution (K_2CrO_4) (Yellow precipitate is formed) (ii) Potassium Iodide (KI) solution. (Yellow precipitate which disappears on warming)
Zn^{2+}	(i) Potassium hexacyanoferrate (II)solution followed by few drops of Dilute HCl. (Bluish white precipitate insoluble in dilute HCl) (ii) Dilute NaOH. (white precipitate soluble in excess)
NH_4^+	Sodium hydroxide solution and warm gently. (Reddish-brown precipitate is formed)
Na^+	Flame test (yellow flame)
Cu^{2+}	(i)Excess ammonia solution (NH_4OH). (Pale blue precipitate soluble in excess of aqueous ammonia forming deep blue solution.) (ii) Potassium hexacyanoferrate (II). (Reddish-brown precipitate.)
Fe^{2+} and Fe^{3+}	(i)Potassium hexacyanoferrate (III) for Fe^{2+} (Deep blue precipitate) (ii) Potassium hexacyanoferrate (II) for Fe^{3+} (Deep blue precipitate)

Uses of Ammonia

❖ Manufacture of fertilizers

Ammonia is used in the manufacture of nitrogenous fertilizers such as ammonium sulphate nitrate (ASN), ammonium sulphate (AS), di-ammonium phosphate (DAP), calcium ammonium nitrate (CAN) and ammonium nitrate (AN).

❖ **Cleaning**

Ammonia softens water and neutralizes acid stains caused by perspiration, thus making washing easier.

❖ **Refrigeration**

Liquid ammonia can be used as a refrigerant because it is highly volatile.

❖ **Manufacture of nitric acid**

Nitric acid is manufactured by the catalytic oxidation of ammonia

Uses of Calcium and its Compounds.

- ❖ used as deoxidizer for steel casting
- ❖ used in extraction of Thorium, Iron and Sodium
- ❖ used for removing last traces of moisture from alcohol
- ❖ slaked lime is used as a building material
- ❖ used in tanning industry
- ❖ calcium carbonate is used in extraction of metals
- ❖ used as a fertilizer e.g calcium sulphate
- ❖ used in the manufacture of cement e.g calcium sulphate

Uses of Sodium and Its Compounds.

- ❖ Used in the manufacture of sodium cyanide (NaCN) which is used in extraction of gold
- ❖ Used in the withdraw of heat from nuclear reactors
- ❖ Used in the manufacture of washing soda (Na_2CO_3)
- ❖ Used in the manufacture of glass (Na_2CO_3)
- ❖ Used for softening water (Na_2CO_3)
- ❖ NaCl used as ingredients in our food in our food
- ❖ Used in food preservation eg NaCl

Uses of Copper and its compounds.

- ❖ Used in the manufacture of electric wires, cables and domestic utensils.
- ❖ Used in preparation of useful alloys such as brass, bronze etc.
- ❖ Used in making coins, jewellery and decorative objects.
- ❖ Used in electroplating

Uses of Iron and its compounds.

- ❖ Used in automobile bodies.
- ❖ Used in making gutter pipes and railway sleepers.
- ❖ Used in making articles which are to stand severe and sudden stress, such as wire ropes bolts chains and agricultural appliances.
- ❖ Used in electromagnetism.
- ❖ Used in making high-speed cutting tools.

Uses of Zinc and its compounds.

- ❖ Used in galvanization of iron.
- ❖ Used as reducing agent.
- ❖ Used for making alloys such as brass.
- ❖ Used in metallurgy of silver and gold.
- ❖ Used as anodic material in dry cells.

Uses of Lead and its compounds.

- ❖ Used in manufacture of electric accumulators.
- ❖ Used in making various alloys.
- ❖ Used in paints (Pb_3O_4).
- ❖ Used for making water pipes.
- ❖ Used in lead storage batteries.
- ❖ Lead compounds are used in preparing high refractive index glasses and in ceramic glasses.
- ❖ Used in the manufacture of bullets, shots etc.

NECTA 2014 CHEMISTRY 2B

3. Sample **J** contains one cation and one anion. Using systematic qualitative analysis procedures carry out the experiment and record carefully your observations, inferences and finally identify the cation and anion present sample **J**. Record your work in a tabular form as shown in Table 2.

Table 2

S/N	Experiment	Observation	Inference

Conclusion

(i)The cation present in sample **J** is_____.

(ii)The anion present in sample **J** is_____.

(iii)The chemical formula of sample **J** is_____

(iv)The chemical name of sample **J** is _____.

SOLUTION.

S/N	Experiment	Observation	Inference
1	Appearance of sample J (i) colour	Pale or light green	Fe^{2+} may be present
	(ii) texture	Crystalline form	NO_3^- , SO_4^{2-} , Cl^- may be present
2	Action of heat; About 0.5g of solid sample J was transferred into a dry test tube and then heated strongly	Colourless gas evolves which turns filter paper dipped in acidified potassium dichromate solution from yellow to green.	SO_4^{2-} of Zn^{2+} , Fe^{2+} , Cu^{2+} may be present.
3	Action of dilute HCl acid on a solid sample J; About 0.5g of solid sample J was transferred into a dry test tube followed by 3 drops of dilute HCl.	No gas evolves	NO_3^- , SO_4^{2-} , Cl^- may be present
4	Action of concentrated H_2SO_4 on a solid sample; About 0.5g of solid sample J was transferred into a dry test tube followed by addition of 3drops of concentrated H_2SO_4 acid, since there was no reaction on cold water then the contents were warmed gently.	No gas evolves.	SO_4^{2-} may be present.

5	Flame test; Nichrome wire was dipped on concentrated. HCl and then small amount of solid sample J was picked using wet wire then heated on a non-luminous flame.	Yellow (orange) sparks.	Fe^{2+} , Fe^{3+} may be present.
6	Solubility of solid samples; About 0.5g of solid sample J was transferred into a dry test tube followed by addition of distilled water to dissolve solid sample J .	Soluble forming pale green solution.	Fe^{2+} may be present.
7	Action of NaOH solution on a sample solution; To about 1cm^3 of the original sample solution of J , sodium hydroxide was added drop-wise until in excess.	Green precipitate was formed, insoluble in excess which turns brown on standing.	Fe^{2+} May be present
8	Action of NH_3 solution on a sample solution; To about 1cm^3 of the original sample solution of J , ammonia solution was added drop-wise until in excess.	Green precipitate was formed, insoluble in excess	Fe^{2+} May be present
9	Confirmatory test for SO_4^{2-}; About 1cm^3 of the original sample solution of J was transferred in a test-tube and then Barium chloride solution was added followed by dilute HCl.	White precipitate	SO_4^{2-} Confirmed
10	Confirmatory test for Fe^{2+}; To about 1cm^3 of the original sample solution of J , few drops of potassium	Deep blue precipitate.	Fe^{2+} Confirmed

	hexacyanoferrate (III).		
--	-------------------------	--	--

Conclusion

- (i) The cation present in sample **J** was Fe^{2+}
(ii) The anion present in sample **J** was SO_4^{2-}
(iii) The chemical formula of sample **J** was FeSO_4
(iv) The chemical name of sample **J** Iron (II)sulphate

PRACTICAL ACTIVITY

3. Sample **D** contains one cation and one anion. Using systematic qualitative analysis procedures carry out the experiment and record carefully your observations, inferences and finally identify the cation and anion present sample **D**. Record your work in a tabular form as shown in Table 2.

Table 2

S/N	Experiment	Observation	Inference

Conclusion

- (i) The cation present in sample **D** is _____.
(ii) The anion present in sample **D** is _____.

SOLUTION

S/N	Experiment	Observation	Inference
1	Appearance of sample D (i) colour	White	NH_4^+ , Na^+ , Ca^{2+} , Zn^{2+} , Pb^{2+} May be present.
	(ii) texture	Powder form.	CO_3^{2-} , HCO_3^-

2	Action of heat; About 0.5g of solid sample D was transferred into a dry test tube and then heated strongly	Colourless gas evolves, which turns lime water milky and wet litmus paper from blue to red.	$\text{CO}_3^{2-}, \text{HCO}_3^-$ May be present
3	Action of dilute HCl acid on a solid sample; About 0.5g of solid sample D was transferred into a dry test tube followed by 3 drops of dilute HCl.	Efferscence of colourless gas evolves, which turns lime water milky and wet litmus paper from blue to red.	$\text{CO}_3^{2-}, \text{HCO}_3^-$ May be present
4	Action of concentrated H_2SO_4 on a solid sample; About 0.5g of solid sample D was transferred into a dry test tube followed by addition of 3 drops of concentrated H_2SO_4 acid, since there was no reaction on cold water then the contents were warmed gently.	Efferscence of colourless gas evolves, which turns lime water milky and wet litmus paper from blue to red.	$\text{CO}_3^{2-}, \text{HCO}_3^-$ May be present
5	Flame test; Nichrome wire was dipped on concentrated. HCl and then small amount of solid sample D was picked using wet wire then heated on a non-luminous flame.	Brick red flame	Ca^{2+} May be present
6	Solubility of solid samples; About 0.5g of solid sample D was transferred into a dry test tube followed by addition of distilled water to dissolve solid sample D .	Insoluble	CO_3^{2-} of $\text{Ca}^{2+}, \text{Pb}^{2+}, \text{Zn}^{2+}, \text{Fe}^{2+}, \text{Fe}^{3+}, \text{Cu}^{2+}$ May be present

	(iii)NaOH solution to the third portion, drop wise till excess		
	(iv)dilute NH_4OH in small amount then in excess to the fourth portion		
	(v) FeSO_4 solution followed by Conc. H_2SO_4 to the fifth portion		
	(vi) KI solution to the sixth portion, warm then cool the mixture.		

Conclusion

- (i) The cation sample R is____.
- (ii) The anion sample R is____.
- (iii)The chemical formula of sample R is____.
- (iv)Write the equation for the reactions that took place at experiments (b) and (c) (ii).

SOLUTION

S/n	Observation	Inference
(a)	White crystalline salt	NH_4^+ , Na^+ , Ca^{2+} , Zn^{2+} , Pb^{2+} May be present. Or Transition metals Fe^{2+} , Fe^{3+} , Cu^{2+} May be absent.
(b)	Cracking sound with brown gas	NO_3^- of Pb^{2+} May be present
(c)	Insoluble in cold water but soluble in hot water	Cl^- of Pb^{2+} May be present

(i)	Effervescence of colourless gas evolves, which turns lime water milky.	$\text{CO}_3^{2-}, \text{HCO}_3^-$ May be present
(ii)	Evolution of brown fumes, which turn wet litmus paper from blue to red.	NO_3^- May be present
(iii)	White precipitate is formed, soluble in excess.	$\text{Zn}^{2+}, \text{Pb}^{2+}$ May be present
(iv)	White precipitate is formed, insoluble in excess.	Pb^{2+} May be present
(v)	Brown ring is formed at the junction of the liquids.	NO_3^- Confirmed
(vi)	Yellow precipitate which disappears on warming but re-appears on cooling	Pb^{2+} Confirmed

Conclusion

- (i) The cation of sample **R** was Pb^{2+}
- (ii) The anion of sample **R** was NO_3^-
- (iii) The chemical formula of sample **R** was $\text{Pb}(\text{NO}_3)_2$
- (iv) $\text{PbNO}_3 (\text{aq}) + \text{H}_2\text{SO}_4 (\text{aq}) \rightarrow \text{PbSO}_4 (\text{aq}) + \text{NO}_2 (\text{g}) + \text{H}_2\text{O} (\text{l})$

NECTA 2017 CHEMISTRY 2A

3. Sample **Q** is a simple salt containing one cation and one anion. Carefully carry out all the experiments described in the Table 2.

iii. Iron has a high tensile strength so it can be stretched without breaking

iv. Iron dissolves in dilute acids

(d) Two uses of iron (Fe)

- ❖ used in Making steel
- ❖ used in Agriculture
- ❖ used in Food and medicine

**NECTA 2018
CHEMISTRY 2A**

3. Sample **N** is a simple salt containing one cation and one anion. Carefully carry out all the experiments described in the Table 2. Record all your observations and appropriate inferences to identify the cation and anion present in sample **N**.

Table 2

S/n	Experiments	Observation	Inference
(a)	Appearance of the solid sample N		
(b)	Put a spatulaful of sample N in a test tube and heat.		
(c)	Add dilute hydrochloric acid to the solid sample N		
(d)	Perform flame test on the sample N .		
(e)	Dissolve the sample in water. Divide the resulting solution into two portions:		
	(i) Add sodium hydroxide solution to the first portion.		
	(ii) Add few drops of magnesium sulphate solution to the second portion, then warm.		

Conclusion

- (i) The cation is ____ and the anion is _____
- (ii) The sample N is _____.
- (iii) Write chemical equations (with state symbols) to show reactions which took place in the experiments (b) and (c).
- (iv) Describe three uses of sample N

SOLUTION

S/N	Observation	Inference
(a)	White powder	NH_4^+ , Na^+ , Ca^{2+} , Zn^{2+} , Pb^{2+} May be present. Or Transition metals Fe^{2+} , Fe^{3+} , Cu^{2+} may be absent.
(b)	Colourless gas evolves, which turn lime water milky and wet litmus paper from blue to red	CO_3^{2-} , HCO_3^- May be present
(c)	Effervescence of colourless gas evolves, which turns lime water milky and wet litmus paper from blue to red	CO_3^{2-} , HCO_3^- May be present
(d)	Golden yellow flame	Na^+ May be present
(e)	Soluble forming colourless solution	CO_3^{2-} , HCO_3^- , of Na^+ , NH_4^+ May be present
(i)	No precipitate is formed, even on warming	Na^+ May be present
(ii)	White precipitate is formed before warming the contents	CO_3^{2-} Confirmed

Conclusion

- (i) The cation was Na^+ and the anion was CO_3^{2-}

CHAPTER 4.

PREPARATION OF STANDARD ACID SOLUTION AND DILUTION FORMULA

This formula for preparation and dilution of standard solution of acids from concentrated one is just used in calculating the volumes of concentrated acids which require dilution to make standard solution.

$$V_{\text{conc}} = \frac{M_d \times R.M.M \times V_d \times 100}{d_{\text{w}}/W}$$

where by

V_{conc} = volume of concentrated acid

w/W (assay) = weight percent.

R.M.M = relative molecular mass

M_d = molarity of dilute acid solution.

V = volume of dilute solution to be prepared.

If molarity of the concentrated acid is to be calculated, then the following formula is applied.

$$M_c = \frac{w/W \times \text{density} \times 1000}{100 \times R.M.M}$$

Where by M_c = molarity of concentrated solution.

After getting molarity of concentrated acid then to know the volume of dilution/volume of distilled water to be added to get dilute standard solution, the following is also applicable; that is dilution formula.

Dilution.

By Definition: Dilution is the process of lowering concentration by adding more solvent (water). Concentrated solution is called stock solution

Dilution formula

Mole remain constant even if solution concentration lowered

Mathematically

Number of moles of concentrated solution (n_c) = Number of moles of dilute solution (n_d)

$$n_c = n_d$$

But:

$$n_c = M_c V_c \quad \text{and} \quad n_d = M_d V_d$$

$$\text{If: } n_c = n_d = M_c V_c = M_d V_d$$

$$\text{Then: } M_c V_c = M_d V_d$$

Where by:

n_c = Number of moles of concentrated solution

n_d = Number of moles of diluted solution

M_c = Molarity of concentrated solution

V_c = volume of concentrated solution

M_d = Molarity of dilute solution

V_d = volume of dilute solution

Dilution Factor.

By Definition: Dilution Factor is the factor shows how times the volume of concentrated solution is diluted to obtain the diluted solution.

Mathematically

From: $M_c V_c = M_d V_d$

Divide by V_d and M_c both sides

$M_c V_c = M_d V_d = k$

Where:

k = diluted factor

$$M_c V_c = M_d V_d$$

Where;

M_c = molarity of concentrated acid.

V_c = volume of concentrated solution.

M_d = molarity of dilute solution.

V_d = volume of dilute solution.

PROBLEM

A solution of concentrated nitric acid, HNO_3 contains 70% by weight of HNO_3 . if the density of the acid of the acid solution is 1.4g/cm^3 . calculate,

(a) the molarity of the solution

(b) the volume of the concentrated acid that will be required to make 1 litre of 0.5M of HNO_3 .

Solution.

Data given.

$w/W(\text{assay}) = 70\%$

density = 1.4

R.M.M = 63 .

$V_d = 1\text{L}$

$M_d = 0.5\text{M}$

$$(a) \text{ molarity of conc solution (M}_c\text{)} = \frac{w/W \times \text{density} \times 1000}{100 \times \text{R.M.M}}$$

$$\text{M}_c = \frac{70 \times 1.4 \times 1000}{100 \times 63} = 15.60$$

Molarity of concentrated solution = 15.60M

(b) from dilution formula = $\text{M}_c \text{XV}_c = \text{M}_d \text{XV}_d$

$$\text{V}_c = \frac{\text{M}_d \text{XV}_d}{\text{M}_c}$$

$$\text{V}_c = \frac{0.5 \times 1}{15.60} = 0.032\text{L}$$

volume concentrated acid = 0.032L = 32cm³.

Problem.

A bottle containing concentrated hydrochloric acid was labeled.

Specific gravity is 1.18

Composition by weight w/W is 36%

(a) what is molarity of this solution?

(b) what volume of the solution.

Solution

(a) molarity of acid.

From the formula.

$$\text{Molarity concentrated acid} = \frac{36 \times 1.18 \times 1000}{100 \times 35.6} = 11.64\text{M}$$

Therefore, Molarity of concentrated acid is 11.64M

(b) by using dilution formula.

$$\text{M}_c \text{XV}_c = \text{M}_d \text{V}_d$$

Where

$$\text{M}_c = 11.64\text{M}$$

$$\text{M}_d = 0.1\text{M}$$

$$\text{V}_c = ?$$

$$\text{V}_d = 1\text{litre}$$

$$\text{V}_c = \frac{0.1 \times 1}{11.64} = 0.00859\text{L} = 8.59\text{cm}^3$$

Volume of a concentrated acid = 0.00859L = 8.59cm³

APPENDIX 1:

Atomic numbers and relative atomic masses of some elements

Element	symbol	Atomic number	Relative atomic mass
Barium	Ba	56	137
Chromium	Cr	24	52
Copper	Cu	29	64
vanadium	V	23	51
Zinc	Zn	30	65
Iron	Fe	26	56
Manganese	Mn	25	55
Silver	Ag	47	108
Lead	Pb	82	207
Iodine	I	53	127

Appendix 2: Periodic table of the elements

Periodic Table of the Elements																						
Period	Group																					
	1																	18				
1	¹ H 1.008 hydrogen																	² He 4.003 helium				
2	³ Li 6.94 lithium	⁴ Be 9.012 beryllium															⁵ B 10.81 boron	⁶ C 12.01 carbon	⁷ N 14.01 nitrogen	⁸ O 16.00 oxygen	⁹ F 19.00 fluorine	¹⁰ Ne 20.18 neon
3	¹¹ Na 22.99 sodium	¹² Mg 24.31 magnesium															¹³ Al 26.98 aluminum	¹⁴ Si 28.09 silicon	¹⁵ P 30.97 phosphorus	¹⁶ S 32.06 sulfur	¹⁷ Cl 35.45 chlorine	¹⁸ Ar 39.95 argon
4	¹⁹ K 39.10 potassium	²⁰ Ca 40.08 calcium	²¹ Sc 44.96 scandium	²² Ti 47.87 titanium	²³ V 50.94 vanadium	²⁴ Cr 52.00 chromium	²⁵ Mn 54.94 manganese	²⁶ Fe 55.85 iron	²⁷ Co 58.93 cobalt	²⁸ Ni 58.69 nickel	²⁹ Cu 63.55 copper	³⁰ Zn 65.38 zinc	³¹ Ga 69.72 gallium	³² Ge 72.63 germanium	³³ As 74.92 arsenic	³⁴ Se 78.97 selenium	³⁵ Br 79.90 bromine	³⁶ Kr 83.80 krypton				
5	³⁷ Rb 85.47 rubidium	³⁸ Sr 87.62 strontium	³⁹ Y 88.91 yttrium	⁴⁰ Zr 91.22 zirconium	⁴¹ Nb 92.91 niobium	⁴² Mo 95.95 molybdenum	⁴³ Tc [97] technetium	⁴⁴ Ru 101.1 ruthenium	⁴⁵ Rh 102.9 rhodium	⁴⁶ Pd 106.4 palladium	⁴⁷ Ag 107.9 silver	⁴⁸ Cd 112.4 cadmium	⁴⁹ In 114.8 indium	⁵⁰ Sn 118.7 tin	⁵¹ Sb 121.8 antimony	⁵² Te 127.6 tellurium	⁵³ I 126.9 iodine	⁵⁴ Xe 131.3 xenon				
6	⁵⁵ Cs 132.9 cesium	⁵⁶ Ba 137.3 barium	⁵⁷⁻⁷¹ La-Lu * lanthanide series	⁷² Hf 178.5 hafnium	⁷³ Ta 180.9 tantalum	⁷⁴ W 183.8 tungsten	⁷⁵ Re 186.2 rhenium	⁷⁶ Os 190.2 osmium	⁷⁷ Ir 192.2 iridium	⁷⁸ Pt 195.1 platinum	⁷⁹ Au 197.0 gold	⁸⁰ Hg 200.6 mercury	⁸¹ Tl 204.4 thallium	⁸² Pb 207.2 lead	⁸³ Bi 209.0 bismuth	⁸⁴ Po [209] polonium	⁸⁵ At [210] astatine	⁸⁶ Rn [222] radon				
7	⁸⁷ Fr [223] francium	⁸⁸ Ra [226] radium	⁸⁹⁻¹⁰³ Ac-Lr ** actinide series	¹⁰⁴ Rf [267] rutherfordium	¹⁰⁵ Db [270] dubnium	¹⁰⁶ Sg [271] seaborgium	¹⁰⁷ Bh [270] bohrium	¹⁰⁸ Hs [277] hassium	¹⁰⁹ Mt [276] meitnerium	¹¹⁰ Ds [281] darmstadtium	¹¹¹ Rg [282] roentgenium	¹¹² Cn [285] copernicium	¹¹³ Nh [285] nihonium	¹¹⁴ Fl [289] flerovium	¹¹⁵ Uup [288] ununpentium	¹¹⁶ Lv [293] livermorium	¹¹⁷ Uus [294] ununseptium	¹¹⁸ Uuo [294] ununoctium				
Atomic number → 1 ← Symbol 1.008 ← Atomic mass ← Name hydrogen																						
Color Code ☐ Metal Solid ☐ Metalloid Liquid ☐ Nonmetal Gas																						

Reference books

1. Lambert,J. and Muir T.A (1973),practical chemistry, Heineman Educational Books, Golborne Lanes,3rd Ed.
2. Chemistry for secondary schools, form 3,south Carolina university & ministry of education and vocational training,first edition,2008
3. Holderness & Lambert,J. New certificate chemistry,6th Ed (1987)
4. Bhushan,B.(1989),Laboratory manual part II, Arya Book Depot, New delh.
5. Hood M.Swaib,(2014),fundamentals of chemistry form 3&4,first edition.
6. Dr S.K. Jain (S. Chand),(2014),Conceptual chemistry, class Xi &Xii.

Welcome at **LUKAGO** Tuition Centre near **Tumbi** secondary
Author is currently teaching Biology and Chemistry at **Tumbi**
secondary located at **Kibaha Education Centre**

Contacts; **+255765776922,**
+255625605414

OTHER BOOKS OF LUKAGO, D

- BIOLOGY FORM 1&2 QUESTIONS AND ANSWERS
- BIOLOGY FORM 3&4 QUESTIONS AND ANSWERS
- BIOLOGY PRACTICAL FOR ORDINARY LEVEL
- CHEMISTRY FORM 1& 2 QUESTIONS AND ANSWERS

LUKAGO, D PAMPHLETS

- ORGANIC CHEMISTRY (FORM 5&6)
- PHYSICAL CHEMISTRY (FORM 5&6)
- GENERAL CHEMISTRY FORM 5
- INORGANIC CHEMISTRY

A comprehensive approach to Ordinary Level chemistry Practical

This is the Highly Successful pamphlet for Ordinary Level Students. The pamphlet Provides Comprehensive Coverage in all Major areas of chemistry practical.



Lukago, D is a graduate of University of Dar es salaam. He holds Bachelor degree of Science with Education (WITH

HONOURS) majoring in Chemistry and Biology (BSc-Ed). He was educated at Nkasi High School and Kabasa Secondary School.

Lukago, D is Currently teaching biology and Chemistry (theory and practical) on both subject at Tumbi Secondary School located at Kibaha Education Centre.



ISBN:9798600861749